

Literature Mining Report

Query Statistics

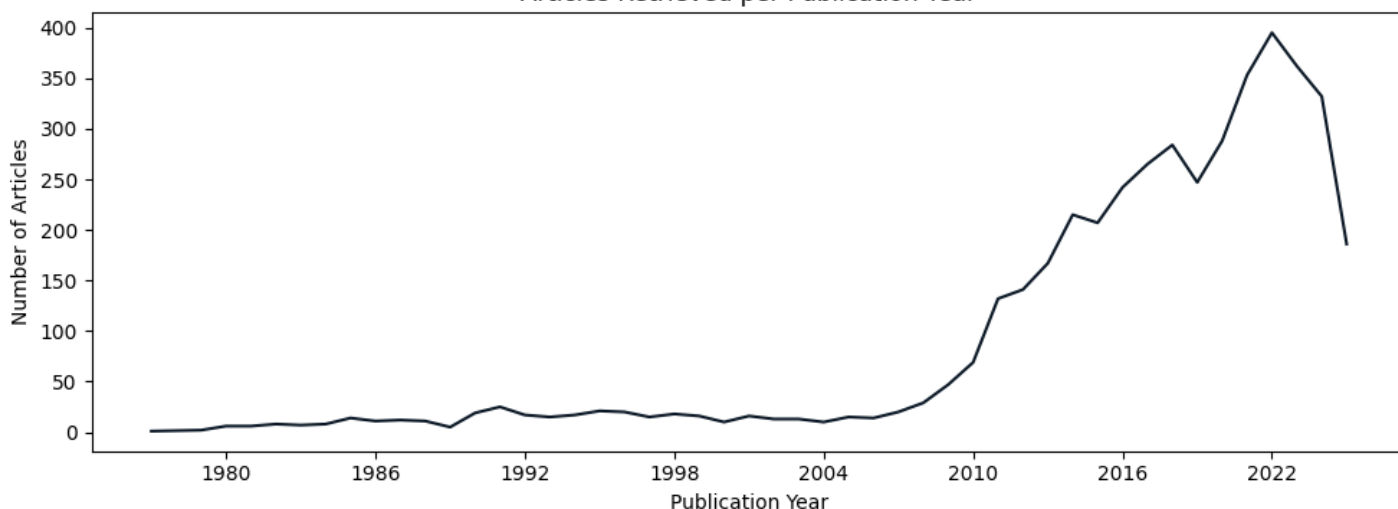
Query date: 2025-08-25

Query terms: "cellulase production trichoderma reesei"

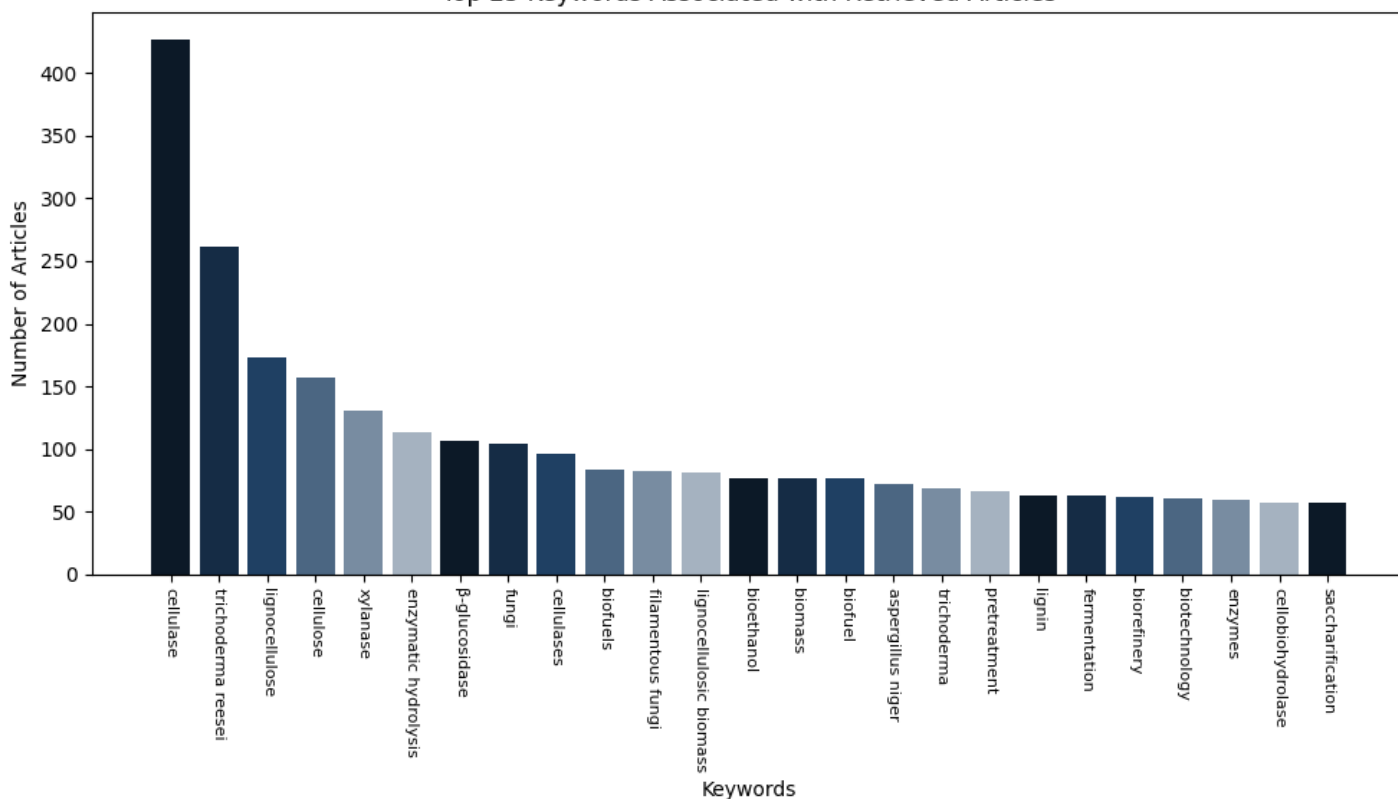
Articles parsed/hits: 4354/4354 (100.00%)

Scoring keywords: cellulase, reesei, cbh1, xyr1 (Default Mode)

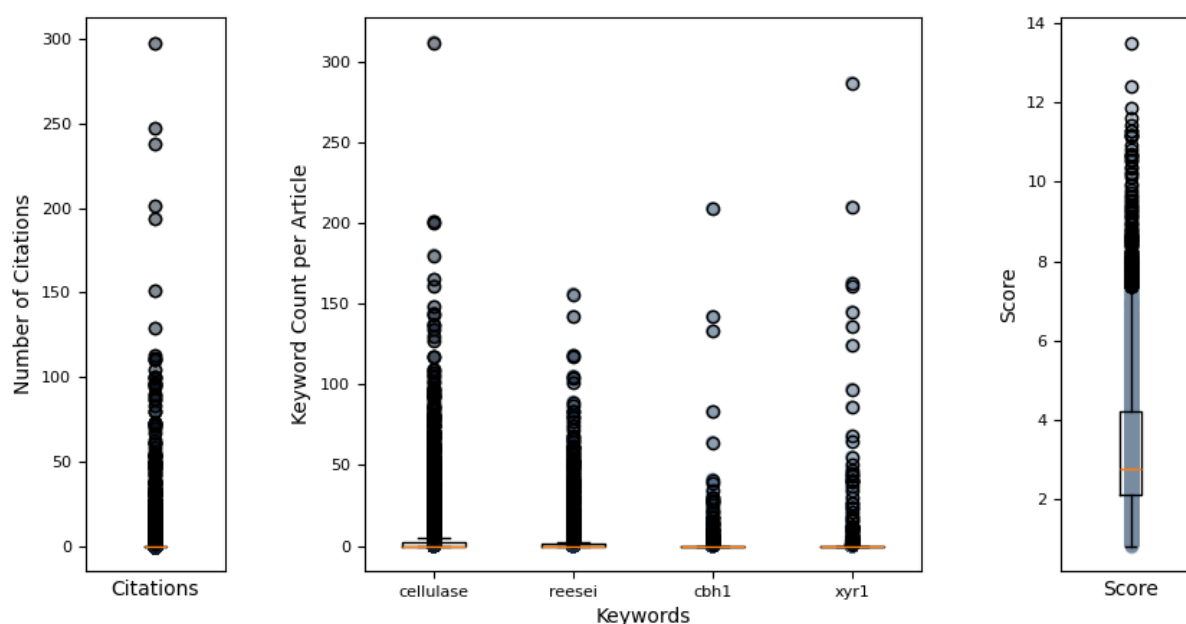
Articles Retrieved per Publication Year



Top 25 Keywords Associated with Retrieved Articles



Hermes Report Summary



Query Results

PMID	Year	Journal	Authors	Title	Citations	Score
22554051	2012	Molecular Micro	Seiboth, Bernha et al.	the putative protein methyltransferase lae1 c...	100	13.51
21526192	2011	PLoS ONE	Suen, Garret et al.	the complete genome sequence of fibrobacter...	97	12.43
33319112	2020	Heliyon	Sulyman, A.O. et al.	isolation, purification and characterization ...	19	11.87
30364340	2018	Frontiers in Bi	Fitz, Elisabeth et al.	the promoter toolbox for recombinant gene exp...	26	11.63
33606681	2021	PLoS Genetics	Wang, Lei et al.	interdependent recruitment of cyc8/tup1 and t...	10	11.43
26583519	2015	Current opinion	Cragg, Simon M et al.	lignocellulose degradation mechanisms across ...	194	11.29
34983541	2022	Microbial Cell	Wang, Yifan et al.	development of a powerful synthetic hybrid pr...	6	11.27
37405433	2023	Applied Microbi	Liu, Dujuan et al.	heterologous protein production in filamentou...	17	11.19
30297963	2018	Scientific Repo	de Paula, Renat et al.	the duality of the mapk signaling pathway in ...	18	11.19
36732590	2023	Scientific Repo	Schalamun, Miri et al.	mapkinases regulate secondary metabolism, sex...	11	11.18
25626518	2015	Molecular Micro	Ghassemi, Sara et al.	the -importin kap8 (pse1/kap121) is required...	15	11.12
23152805	2012	PLoS ONE	Zhang, Jiwei et al.	ras gtpases modulate morphogenesis, sporulati...	16	10.93
38548869	2024	Communications	Zhu, Zhihua et al.	the protein methyltransferase trsam inhibits ...	4	10.80
31849916	2019	Frontiers in Mi	Qian, Yuanchao et al.	enhancement of cellulase production in tricho...	14	10.72
27150814	2017	Fungal Genetics	Daly, Paul et al.	transcriptomic responses of mixed cultures of...	12	10.69
28302998	2017	Scientific Repo	Berlemont, Rena	distribution and diversity of enzymes for pol...	39	10.67
35852347	2022	Microbiology Sp	Wang, Zhixing et al.	functional characterization of sugar transpor...	7	10.64
18471273	2008	Biotechnology f	Olofsson, Kim et al.	a short review on ssf an interesting proces...	111	10.55
24795706	2014	Frontiers in Mi	Kricka, William et al.	metabolic engineering of yeasts by heterologo...	22	10.51
35910619	2022	Frontiers in Mi	Zhang, Guoyan et al.	design and application of an efficient cellul...	9	10.39
20466064	2010	Fungal Genetics	Schmoll, Monika et al.	trichoderma in the light of day physiology ...	54	10.35
35077963	2022	Ultrasonics Son	Li, Hongyan et al.	the promoted hydrolysis effect of cellulase w...	5	10.22
39629327	2024	Engineering Mic	Li, Zhongye et al.	research advances on the consolidated bioproc...	5	10.20
32963975	2020	Biotechnology R	Chettri, Dixita et al.	innovations in cazyme gene diversity and its ...	22	10.14
34059751	2021	Communications	Tu, Ran et al.	droplet-based microfluidic platform for high...	20	9.93

Query Result 1/25

The putative protein methyltransferase LAE1 controls cellulase gene expression in *Trichoderma reesei*

Seiboth, Bernhard; Karimi, Razieh Aghchah; Phatale, Pallavi A; Linke, Rita; Hartl, Lukas; Sauer, Dominik G; Smith, Kristina M; Baker, Scott E; Freitag, Michael; Kubicek, Christian P

Molecular Microbiology

Year: 2012 PMID: 22554051 doi: 10.1111/j.1365-2958.2012.08083.x Citations: 100 Score: 13.51

Keywords hits: cellulase: 89, reesei: 80, cbh1: 0, xyr1: 24

Abstract:

Trichoderma reesei is an industrial producer of enzymes that degrade lignocellulosic polysaccharides to soluble monomers, which can be fermented to biofuels. Here we show that the expression of genes for lignocellulose degradation are controlled by the orthologous *T. reesei* protein methyltransferase LAE1. In a *lae1* deletion mutant we observed a complete loss of expression of all seven cellulases, auxiliary factors for cellulose degradation, -glucosidases and xylanases were no longer expressed. Conversely, enhanced expression of *lae1* resulted in significantly increased cellulase gene transcription. *Lae1*-modulated cellulase gene expression was dependent on the function of the general cellulase regulator XYR1, but also *xyr1* expression was LAE1-dependent. LAE1 was also essential for conidiation of *T. reesei*. Chromatin immunoprecipitation followed by high-throughput sequencing (ChIP-seq) showed that *lae1* expression was not obviously correlated with H3K4 di- or trimethylation (indicative of active transcription) or H3K9 trimethylation (typical for heterochromatin regions) in CAZyme coding regions, suggesting that LAE1 does not affect CAZyme gene expression by directly modulating H3K4 or H3K9 methylation. Our data demonstrate that the putative protein methyltransferase LAE1 is essential for cellulase gene expression in *T. reesei* through mechanisms that remain to be identified.

Summary:

To test whether this was caused by a loss of cellulase production, we cultivated the *lae1* deletion strains and the parent strain on lactose, a carbon source that induces cellulase expression, but whose utilization is independent of the action of secreted cellulases. Interestingly, 13 of these 28 clusters were found in areas not previously predicted as CAZyme gene clusters. In order to confirm the results from microarray analysis, we used *cel7A* and *cel6A* as cellulase model genes and tested their expression in the parent strain and in the *lae1* strain by qRT-PCR. The results also demonstrate that cellulase regulation by LAE1 is independent of the nature of the inducer. Having identified LAE1 as a regulator of cellulase and hemicellulase biosynthesis in *T. reesei*, we hypothesized that its activity in the parent strain could be limiting for cellulase gene expression. We must also note that although ChIP analysis of heterochromatic marks in *A. nidulans laeA* strains had revealed a dramatic increase in H3K9me3 and HepA binding at secondary metabolism clusters a direct effect of *LaeA* on histone modification still awaits biochemical evidence. While the precise mechanism of action of LAE1 remains to be identified, we speculate that it could be related to the linkage between asexual sporulation and CAZyme gene transcription in *T. reesei*: as we have shown in this article, one of the most striking phenotypes of *lae1*-mutants in *T. reesei* is the almost complete absence of sporulation in the *lae1* and hypersporulation in *tef1: lae1* strains, a phenomenon already observed in some other fungi. Asexual sporulation triggers massive CAZyme gene expression in *T. reesei* in an inducer-independent but XYR1-dependent way, and sporulation is commonly observed at later stages of cellulase formation on lactose which was absent from the *lae1* strain and increased in the *tef1: lae1* mutant in this study.

Figures:

Figure 1: Phylogenetic analysis of *LaeA/LAE1* proteins from Eurotiomycetes, Dothidiomycetes and Sordariomycetes.

Figure 2: Effect of loss-of-function of *lae1* on biomass formation and cellulase/hemicellulase enzyme formation by *T. reesei*.

Figure 3: Growth of *T. reesei* QM 9414 (blue), the *lae1* strain (CPK3793, red) and the strain expressing *tef1:lae1* (CPK4086, green) on several oligosaccharides.

Figure 4: Expression of the two cellobiohydrolase-encoding genes *cel7a* and *cel6a* in *T. reesei* QM 9414 and the *lae1* strain CPK3793 during growth on lactose or incubation with sophorose.

Figure 5: Biomass formation (A), cellulase production (B) and extracellular protein (C) during growth of *T. reesei* QM 9414 (QM) and several mutant strains bearing additional copies of the *tef1:lae1* gene construct (CPK3791, CPK4326, CPK4086, CPK4087, CPK4325) on lactose.

Figure 6: Relative abundance of transcripts for *cel7a*, *cel6a* and *lae1* at 26 h of growth on lactose in two *T. reesei tef1:lae1* mutant strains (CPK4086, grey bars; CPK3791, white bars) in relation to the QM 9414 recipient strain (black bars).

Figure 7: Cellulase activity during growth of selected *T. reesei tef1:lae1* mutant strains on cellulose.

Figure 8: Relative changes in expression of CAZyme genes in the *lae1* (CPK3793) and the *tef1:lae1* (CPK4086) strains, given as the fold changes of transcript hybridization in relation to the parent strain.

Figure 9: Cellulase formation in delta- *lae1* (CPK3793) and delta- *xyr1* strains of *T. reesei* is unaffected by the constitutive overexpression of *xyr1* or *lae1*.

Figure 10: Selected results of ChIP-seq with antibodies against H3K9me3, H3K4me3 and H3K4me2 in *lae1* (CPK3793), wild type (QM 9414) and the LAE1-overexpressing *tef1:lae1* strain CPK4086.

Figure 11: LAE1 affects sporulation in *T. reesei*.

Mentioned biomedical entities:

Genes: CIP1, Cellulase, GH, HP1, HepA, LAE1, LaeA, XYN1, XYN2, XYN3, XYR1, cel12A, cel1A, cel5A, cel6A, cel7B, cellulase, cip1, histone, lae1, pki1, tef1, xyn1, xyn3, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: *A. fumigatus*

Tissues: N/A

Pathways: N/A

Query Result 2/25

The Complete Genome Sequence of Fibrobacter, bsucc: bogenes S85 Reveals a Cellulolytic and Metabolic, bogenes S85 Genome

Suen, Garret; Weimer, Paul J.; Stevenson, David M.; Aylward, Frank O.; Boyum, Julie; Deneke, Jan; Drinkwater, Colleen; Ivanova, Natalia N.; Mikhailova, Natalia; Chertkov, Olga; Goodwin, Lynne A.; Currie, Cameron R.; Mead, David; Brumm, Phillip J.; Li, Wenjun
 PLoS ONE

Year: 2011 PMID: 21526192 doi: 10.1371/journal.pone.0018814 Citations: 97 Score: 12.43

Keywords hits: cellulase: 96, reesei: 1, cbh1: 0, xyr1: 0

Abstract:

, bFibrobacter succinogenes is an important member of the rumen, bc toggle=yesF. succinogenes, we sequenced the genome of the type strain, bc toggle=yesClostridium thermocellum or the ruminococci 12. In F., bc toggle=yesFibrobacter succinogenes) to completion. A metabolic, bc toggle=yesF. succinogenes was originally placed within the phylum, bc toggle=yesF. succinogenes belongs to its own phylum.

Summary:

Herbivorous mammals are essential components of terrestrial ecosystems and are major, bhe cell, and further lacks a xylose isomerase to convert, bheir environmental niche. Microb Ecoldoi:10.1007/s00248-010-9692-810.1007/s00248-010-9692-820585943, btuents)0.1328.6 GlucomannanSalep Mixed -1,4-Glc and Man0.4403.9 XyloglucanTamarind-1,4-Glc 0259.5 PectinCitrus-1,4-GalUA 0.1167.7 InulinChicory-2,1-Fru1.3532 Phlein Orchardgrass-2,6-Fru0.313.8, bates. The function of these enzymes may be to provide growth substrates, btuted xylans. Alternately, the organism may utilize an unrecognized gene, bates mainly to glucose, and its adjacent upstream homolog from, breco: btruction analysis of this genome reveals how F. succinogenes, bred. Comparison of the predicted proteins, btruction analysis using the software program PRIAM to generate KEGG, bis p: bsiologically adapted to the rumen environment. An analysis of its plant, bhas: be ability to hydrolyze a wide variety of polysaccharides, it can only utilize, ben demonstrated in cell-free extracts.

Figures:

Figure 1: An overview of metabolism and transport inFibrobacte, bThe: bjour fermentative products succinate, acetate, and formate are, bmbination of adherence molecules, carbohydrate-binding molecules, and, bpleted genome sequence ofF. succinogenesS85 is, bserved activities of the cloned proteins are shown inTable, bPred: bted transporters are also shown, including sodium ion channel, bof s: butes is shown through the direction of the arrow through the, bsolu: bs transported by channel proteins; secondary transporters with two, be cellulase activity in fermentations actively degrading cellulose7; ii) has an, be cellulases and our genomic analysis reveals no homologs to known, barro: b into the cell indicating the solute and coupling ion; ATP-driven, bunkn: bn energy-coupling mechanism, shown with a single arrow. Some, bmult: bstep pathways are not fully-represented, and are denoted with, boran: b arrowheads. Abbreviations: AA, amino acids; Ala, alanine; Met,, bmeth: bnine; Pro, proline; PRPP,, b5-phospho--D-ribose 1-diphosphate; PEP,, bnot: bve an Entner-Doudoroff pathway and that the glyoxylate shunt is also, bund inR. flavefaciens, whereas a GH48 exoglucanase is, bilize this as an energy source because it lacks both a xylose permease for, bt clear how the genes identified in the organism contribute to its success. It, babse: b. This bacterium also appears to have incomplete pathways for the, b(Table 1).

Mentioned biomedical entities:

Genes: N/A

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: N/A

Tissues: N/A

Pathways: N/A

Query Result 3/25

Isolation, purification and characterization of cellulase produced by *Aspergillus niger* cultured on *Arachis hypogaea* shells

Sulyman, A.O.; Igunnu, A.; Malomo, S.O.

Heliyon

Year: 2020 PMID: 33319112 doi: 10.1016/j.heliyon.2020.e05668 Citations: 19 Score: 11.87

Keywords hits: cellulase: 161, reesei: 3, cbh1: 0, xyr1: 0

Abstract:

Cellulases are enzymes that hydrolyse cellulose and related cellu-oligosaccharides derivatives. Its applications are enormous but high cost of production is the bottle-neck against the utilization of cellulase in industries. Therefore, this study investigated the isolation, purification and characterization of cellulase produced by *Aspergillus niger* cultured on *Arachis hypogaea* shells. The crude cellulase enzyme was produced by *A.niger* through submerged fermentation process using *A.hypogaea* shells as a carbon source. The optima fermentation conditions were determined by varying different parameters. The crude cellulase was purified through ammonium sulphate precipitation, dialysis and gel-filtration chromatography. The molecular weight was estimated using sodium dodecyl sulphate polyacrylamide gel electrophoresis. The effects of pH and temperature on the activity of the purified cellulase were investigated. The study revealed that the: optimal production of crude cellulase was achieved at incubation period of 120 h, pH 4, temperature 40 C, and inoculum size of 13 105 CFU/ml. Cellulase was purified to 68.12-fold with a yield and specific activity of 3.87% and 484.3 U/mg respectively. The Vmax for the cellulase was 9.26 U/ml while the Km was 0.23 mg/ml. The molecular weight of the cellulase was approximately 13.5 kDa and the enzyme has higher specificity for CMC compared to other substrates. The optimum pH and temperature for the cellulase activity were 4 and 40 C respectively. The present study has shown that *A.hypogaea* shells can be used as a carbon source by *A.niger* for the production of cellulase.

Summary:

The high crude fibre content obtained from this study for untreated, pretreated shells is an indication that *A. hypogaea* shell is very rich in cellulose and can serve as an inducer for cellulase production. The differences in the molecular weight of purified cellulase obtained from this study and those earlier reported may be as a result of different species of microorganisms as well as agricultural wastes used as substrate in this study. 4.6 Km is a measure of affinity a particular enzyme has for its substrates. The value obtained for Kcat, from this study is 0.08 s⁻¹ which implies that more enzyme will be needed to convert substrate to product. 4.7 The effect of pH on the activity of purified cellulase was presented in. The optimum enzyme activity was observed at pH 4.0 in this study. Temperature optimum for purified cellulase was observed at 40 C.

Figures:

Figure 1: Cellulase activity of *A.niger* cultured on *A.hypogaea* shell at varying incubation times.

Figure 2: Cellulase activity of *A.niger* cultured on *A.hypogaea* shell at varying pH.

Figure 3: Cellulase activity of *A.niger* cultured on *A.hypogaea* shell at varying temperatures.

Figure 4: Cellulase activity of *A.niger* cultured on *A.hypogaea* shell at varying substrate concentrations.

Figure 5: Cellulase activities produced by *A.niger* cultured on *A.hypogaea* shell at varying inoculum size.

Figure 6: Cellulase activities produced by *A.niger* cultured on *A.hypogaea* shell using different nitrogen source.

Figure 7: Elution Profile of *A.hypogaea* shell cellulase on sephadex G-100 chromatography.

Figure 8: SDS-PAGE profile of cellulase produced by *A.niger* cultured on treated *A.hypogaea* shell.

Figure 9: Substrate kinetics of purified cellulase produced by *A.niger*.

Figure 10: Lineweaver-Burk plot of cellulase hydrolyzing carboxymethylcellulose.

Figure 11: Effect of pH on endoglucanase activity of purified cellulase produced by *A.niger*.

Figure 12: Effect of temperature on endoglucanase activity of purified cellulase produced by *A.niger*.

Mentioned biomedical entities:

Genes: B, Cellulase, albumin, cellulase, lipase

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: *Arachis hypogaea*

Tissues: pulp, tube

Pathways: N/A

Query Result 4/25**The Promoter Toolbox for Recombinant Gene Expression in *Trichoderma reesei***

Fitz, Elisabeth; Wanka, Franziska; Seiboth, Bernhard

Frontiers in Bioengineering and Biotechnology

Year: 2018

PMID: 30364340

doi: 10.3389/fbioe.2018.00135

Citations: 26

Score: 11.63

Keywords hits: cellulase: 60, reesei: 54, cbh1: 2, xyr1: 9

Abstract:

The ascomycete *Trichoderma reesei* is one of the main fungal producers of cellulases and xylanases based on its high production capacity. Its enzymes are applied in food, feed, and textile industry or in lignocellulose hydrolysis in biofuel and biorefinery industry. Over the last years, the demand to expand the molecular toolbox for *T. reesei* to facilitate genetic engineering and improve the production of heterologous proteins grew. An important instrument to modify the expression of key genes are promoters to initiate and control their transcription. To date, the most commonly used promoter for *T. reesei* is the strong inducible promoter of the main cellobiohydrolase *cel7a*. Beside this one, there is a number of alternative inducible promoters derived from other cellulase- and xylanase encoding genes and a few constitutive promoters. With the advances in genomics and transcriptomics the identification of new constitutive and tunable promoters with different expression strength was simplified. In this review, we will discuss new developments in the field of promoters and compare their advantages and disadvantages. Synthetic expression systems constitute a new option to control gene expression and build up complex gene circuits. Therefore, we will address common structural features of promoters and describe options for promoter engineering and synthetic design of promoters. The availability of well-characterized gene expression control tools is essential for the analysis of gene function, detection of bottlenecks in gene networks and yield increase for biotechnology applications.

Summary:

In the cellulase hyperproducer strain *T. reesei* Rut-C30, ancestor of many industrial strains, carbon catabolite repression was abolished and therefore the strain can express cellulase and xylanase genes on d-glucose and other non-inducing carbon sources. This repressible promoter is not restricted to wheat straw cultures and was also functional during growth on other carbon sources including d-glucose and glycerol. Analogous to l-methionine repressible promoters, new inducible promoters were discovered which do not interfere with cellulase production or growth of *T. reesei*. The system comprises two different expression cassettes, including one which provides a weak constitutive level of a synthetic transcription factor and a second one for strong tunable expression of the target gene via an synthetic transcription factor dependent promoter consisting of eight BM3R1 binding sites. The rational design of promoters is one of the newest fields within synthetic biology. Natural promoter sequences with known expression characteristics or strength can be aligned and similar parts found among the sequences in these promoters can be divided into building blocks. An advantage of this system is that first hints for new transcription factor binding sites and their respective transcription factors can be found without prior knowledge. Although only present in 57% of all eukaryotic promoters, the TATA-box is one of the most known and well characterized core promoter elements.

Figures:

Figure 1: A simplified model for the different mode of actions of constitutive and tunable promoters.

Figure 2: A schematic presentation of synthetic expression system for gene activation.

Figure 3: The different levels of regulation.

Mentioned biomedical entities:

Genes: ACE2, ARA1, CEL7A, CRE1, Cel7a, Cellulase, FW, LexA, MET3, RCE1, TFIIB, XYN1, XYN2, XYN3, XYR1, ace2, bphA, cellulase, cre1, eno1, gpd1, pki1, qa-2, tTA, tef1, thiA, xyn1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: *A. niger*, *Escherichia coli*, Human, human

Tissues: N/A

Pathways: N/A

Query Result 5/25

Interdependent recruitment of CYC8/TUP1 and the transcriptional activator XYR1 at target promoters is required for induced cellulase gene expression in *Trichoderma reesei*

Wang, Lei; Zhang, Weixin; Cao, Yanli; Zheng, Fanglin; Zhao, Guolei; Lv, Xinxing; Meng, Xiangfeng; Liu, Weifeng; Butler, Geraldine
PLoS Genetics

Year: 2021 PMID: 33606681 doi: 10.1371/journal.pgen.1009351 Citations: 10 Score: 11.43

Keywords hits: cellulase: 74, reesei: 33, cbh1: 3, xyr1: 86

Abstract:

Cellulase production in filamentous fungus *Trichoderma reesei* is highly responsive to various environmental cues involving multiple positive and negative regulators. XYR1 (Xylanase regulator 1) has been identified as the key transcriptional activator of cellulase gene expression in *T. reesei*. However, the precise mechanism by which XYR1 achieves transcriptional activation of cellulase genes is still not fully understood. Here, we identified the TrCYC8/TUP1 complex as a novel coactivator for XYR1 in *T. reesei*. CYC8/TUP1 is the first identified transcriptional corepressor complex mediating repression of diverse genes in *Saccharomyces cerevisiae*. Knockdown of *Trcyc8* or *Trtup1* resulted in markedly impaired cellulase gene expression in *T. reesei*. We found that TrCYC8/TUP1 was recruited to cellulase gene promoters upon cellulose induction and this recruitment is dependent on XYR1. We further observed that repressed *Trtup1* or *Trcyc8* expression caused a strong defect in XYR1 occupancy and loss of histone H4 at cellulase gene promoters. The defects in XYR1 binding and transcriptional activation of target genes in *Trtup1* or *Trcyc8* repressed cells could not be overcome by XYR1 overexpression. Our results reveal a novel coactivator function for TrCYC8/TUP1 at the level of activator binding, and suggest a mechanism in which interdependent recruitment of XYR1 and TrCYC8/TUP1 to cellulase gene promoters represents an important regulatory circuit in ensuring the induced cellulase gene expression. These findings thus contribute to unveiling the intricate regulatory mechanism underlying XYR1-mediated cellulase gene activation and also provide an important clue that will help further improve cellulase production by *T. reesei*.

Summary:

Altogether, the data indicate that TrCYC8/TUP1 plays an important role in mediating the induced cellulase gene expression. To test whether TrCYC8/TUP1 is recruited to the cellulase gene promoter and directly involved in activating cellulase gene expression, chromatin immunoprecipitation followed by quantitative PCR on selected promoter regions was performed using TrTUP1 antibody to determine TrTUP1 occupancy on cellulase gene promoters. In all these data suggest that TrCYC8/TUP1 is involved in remodeling nucleosomes positioned in cellulase gene promoters to activate their transcription although more precise regulatory mechanism needs to be investigated. The results of this study show that a TrCYC8/TUP1 complex exists in *T. reesei*, and is required for high-level activation of cellulase genes in vivo. Nonetheless, the impaired cellulase gene expression with repressed *Trcyc8* or *Trtup1* in this study implicates that the complex may act in a different mode from that of CRE1 in controlling the cellulolytic response in *T. reesei* although possible interactions between CRE1 and TrCYC8/TUP1 can not be excluded. Interestingly, the CYC8/TUP1 complex has been also reported to function in gene activation. Our observations that TrCYC8/TUP1 is absolutely required for induced cellulase gene expression and is only recruited to cellulase promoters upon induction indicate that a different mode of activation by TrCYC8/TUP1 is adopted. Nevertheless, there has been no report on CYC8/TUP1 function in gene activation in filamentous fungi although RCM1/RCO1, the CYC8/TUP1 homolog in *Neurospora crassa*, has been reported to mediate specific repression of the frequency gene. Considering the critical role of XYR1 in the induced cellulase gene expression, it is reasonable to attribute the observed defect in transcriptional activation of cellulase genes with repressed *Trtup1* or *Trcyc8* to the impaired XYR1 binding to its target genes.

Figures:

Figure 1: Evolutionarily conserved CYC8/TUP1 complex exists in *T. reesei*.

Figure 2: Knockdown of *Trcyc8* and *Trtup1* by conditional repression.

Figure 3: Knockdown of *Trcyc8* or *Trtup1* compromised *T. reesei* growth and conidiation on agar plates but not in liquid media.

Figure 4: Knockdown of *Trcyc8* and *Trtup1* abolished cellulase production.

Figure 5: *Trcyc8* or *Trtup1* repression resulted in a significant decrease in the transcription of cellulase gene and *xyr1* transcription.

Figure 6: TrTUP1 was recruited to cellulase gene promoters upon induction in an XYR1-dependent manner.

Figure 7: XYR1 overexpression failed to rescue the defective cellulase production in *Ptcu1-Trcyc8KD* or *Ptcu1-Trtup1*.

Figure 8: Efficient binding of XYR1 to cellulase genes in vivo relied on TrTUP1.

Figure 9: The effect of *Trcyc8* or *Trtup1* repression on the association of histone H4 with cellulase gene promoters upon cellulose induction.

Figure 10: A working model for interdependent binding of TrCYC8/TUP1 and XYR1 on cellulase gene promoters to mediate their induced expression.

Mentioned biomedical entities:

Genes: CBH1, CEL7A, CRE1, CYC8, Cellulase, IP, LexA, MA, MIG1, PIC, SAGA, Sspl, TUP1, WD40, XYR1, Xyr1, actin, arg1, cbh1, cellulase, cre1, cyc1, histone, tup1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: Escherichia coli, S. cerevisiae

Tissues: N/A

Pathways: N/A

Query Result 6/25

Lignocellulose degradation mechanisms across the Tree of Life

Cragg, Simon M; Beckham, Gregg T; Bruce, Neil C; Bugg, Timothy DH; Distel, Daniel L; Dupree, Paul; Etxabe, Amaia Green; Goodell, Barry S; Jellison, Jody; McGeehan, John E; McQueen-Mason, Simon J; Schnorr, Kirk; Walton, Paul H; Watts, Joy EM; Zimmer, Martin

Current opinion in chemical biology

Year: 2015 PMID: 26583519 doi: 10.1016/j.cbpa.2015.10.018 Citations: 194 Score: 11.29

Keywords hits: cellulase: 14, reesei: 0, cbh1: 0, xyr1: 0

Abstract:

Organisms use diverse mechanisms involving multiple complementary enzymes, particularly glycoside hydrolases (GHs), to deconstruct lignocellulose. Lytic polysaccharide monooxygenases (LPMOs) produced by bacteria and fungi facilitate deconstruction as does the Fenton chemistry of brown-rot fungi. Lignin depolymerisation is achieved by white-rot fungi and certain bacteria, using peroxidases and laccases. Meta-omics is now revealing the complexity of prokaryotic degradative activity in lignocellulose-rich environments. Protists from termite guts and some oomycetes produce multiple lignocellulolytic enzymes. Lignocellulose-consuming animals secrete some GHs, but most harbour a diverse enzyme-secreting gut microflora in a mutualism that is particularly complex in termites. Shipworms however, house GH-secreting and LPMO-secreting bacteria separate from the site of digestion and the isopod *Limnoria* relies on endogenous enzymes alone. The omics revolution is identifying many novel enzymes and paradigms for biomass deconstruction, but more emphasis on function is required, particularly for enzyme cocktails, in which LPMOs may play an important role.

Summary:

Here we focus on the CAZy categories of glycoside hydrolases and Auxiliary Activities redox enzymes that act with GHs, often in a synergistic manner. The enzymatic degradation of cellulose and hemicellulose is accomplished in Nature via the collective action of multiple carbohydrate-active enzymes, typically acting together as a cocktail with complementary, synergistic activities and modes of action. For example, filamentous cellulolytic fungi produce GH Family 7 enzymes, which are potent cellobiohydrolases, but in prokaryotes this function is provided by other families such as GH48. Some enzymes have interpolation between these paradigms wherein a single protein contains more than one active site, for example, a multimodular enzyme with GH5, GH6, CBM5 and CBM10 domains has been found. Recently, a new oxidative enzymatic paradigm was discovered for cleavage of polysaccharide linkages these enzymes have been termed lytic polysaccharide monooxygenases, but some were originally classified as GH Family 61 cellulases and others, Family 33 Carbohydrate-Binding Modules. Some white rot fungi have also been shown to have large numbers of LPMO genes. Brown rot fungi have evolved multiple times from the predecessors of current white rot fungi and in these evolutionary advances, lignolytic enzyme systems and crucial types of cellulases have been lost. The efficiency of the CMF system is thought to provide brown rot fungi advantages in exploiting ecological niches, and for example, these fungi have displaced white rot predecessors in the degradation of conifer wood. Some ascomycete fungi can also degrade wood cell walls, forming chains of diamond-shaped cavities that generally follow the orientation of the S2 elementary fibrils, causing soft rot.

Figures:

Mentioned biomedical entities:

Genes: CBM5, GH, GHs, U19, cellulase, peroxidase

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: N/A

Tissues: N/A

Pathways: N/A

Query Result 7/25

Development of a powerful synthetic hybrid promoter to improve the cellulase system of *Trichoderma reesei* for efficient saccharification of corncob residues

Wang, Yifan; Liu, Ruiyan; Liu, Hong; Li, Xihai; Shen, Linjing; Zhang, Weican; Song, Xin; Liu, Weifeng; Liu, Xiangmei; Zhong, Yaohua
Microbial Cell Factories

Year: 2022 PMID: 34983541 doi: 10.1186/s12934-021-01727-8 Citations: 6 Score: 11.27

Keywords hits: cellulase: 44, reesei: 61, cbh1: 133, xyr1: 5

Abstract:

The filamentous fungus *Trichoderma reesei* is a widely used workhorse for cellulase production in industry due to its prominent secretion capacity of extracellular cellulolytic enzymes. However, some key components are not always sufficient in this cellulase cocktail, making the conversion of cellulose-based biomass costly on the industrial scale. Development of strong and efficient promoters would enable cellulase cocktail to be optimized for bioconversion of biomass.

Summary:

Therefore, optimization of *T. reesei* cellulase system by overexpressing specific essential enzymes directed by powerful promoters is a promising solution to improve the lignocellulose hydrolysis efficiency. In this study, a novel synthetic hybrid promoter Pcc was constructed by fusing the modified activation region of the strongest inducible promoter Pcbh1 to the core region of a strong constitutive promoter Pcdna1 after identifying their function. Then, the promoters Pcbh1-dc and Pcbh1-cr were fused with egfp to construct the corresponding cassettes, respectively. Based on the observation that the total cellulase activity of *T. reesei* can be improved by overexpressing specific cellulases directed by Pcc, this hybrid promoter was demonstrated to be efficient in optimizing cellulase system. Given that the individual overexpression of BGLA and EG2 driven by Pcc could led to improvement of the activities of BGL and EG, respectively, and both of them resulted in the FPA augmentation, a hypothesis was proposed that construction of strains co-overexpressing BGLA and EG2 directed by Pcc could further optimize the *T. reesei* enzymatic system to improve the cellulolytic ability. Then, QPEB29 and QPEB70 were cultured in CPM for 7 days at 30 C for cellulase production. The selectable marker hygromycin B gene fragment was amplified by hph-F/hph-R. Then the five differently truncated promoters of cdna1 were respectively employed to fuse with the upstream homologous arm of cbh1, hph and cbh1-ORF through Double-joint PCR, and primer pair cbh1-UF/cbh1-R was used to amplify the final cassettes.

Figures:

Figure 1: Nucleotide sequences of the P cdna1 (a) and P cbh1 (b).

Figure 2: Identification of the truncated fragments of P cdna1 fused with cbh1 as the reporter gene.

Figure 3: Identification of P cbh1 activation region (AR) using egfp as the reporter gene.

Figure 4: Construction of hybrid promoter P cc .

Figure 5: Expression of bglA driven by P cc resulted in increase of BGL activity and FPA.

Figure 6: Expression of eg2 driven by P cc resulted in increase of EG activity and FPA.

Figure 7: Co-overexpression of bglA and eg2 driven by P cc in *T. reesei* resulted in increase of FPA.

Figure 8: Saccharification of different pretreated corncob residues using the extracellular enzymes from QP4 and the transformants co-overexpressing bglA and eg2 driven by P cc .

Mentioned biomedical entities:

Genes: AR, Ace1, Ace2, BGL1, BGLA, CBH, CBH1, Cre1, EG2, FPA, Hap5, PBC1, PBD1, PD1, TATA, Xyr1, bgl1, cbh1, cellulase, gpdA, hph, pdc, ptrA

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: *A. niger*, *Penicillium decumbens*

Tissues: N/A

Pathways: N/A

Query Result 8/25**Heterologous protein production in filamentous fungi**

Liu, Dujuan; Garrigues, Sandra; de Vries, Ronald P.

Applied Microbiology and Biotechnology

Year: 2023

PMID: 37405433

doi: 10.1007/s00253-023-12660-8

Citations: 17

Score: 11.19

Keywords hits: cellulase: 7, reesei: 52, cbh1: 15, xyr1: 0

Abstract:

, be

Summary:

In addition, 11 different promoters for heterologous protein-encoding gene expression in *A. vadensis* have been tested and their efficacy has been addressed. There are also other *Aspergillus* species that have been developed into heterologous protein production platforms. The expression of *A.* Subsequently, the levels of production of three heterologous proteins, bacterial xylanase, fungal immunomodulatory proteins, and human serum albumin, were increased in *T. reesei* under the control of *cbh1* promoter. Owing to the high efficiency of protein secretion and excellent capability of posttranslational modification, filamentous fungi are considered as promising cell factories for heterologous protein expression and secretion. Examples of enzymes produced with these and other inducible promoters as well as the production yields achieved are shown in Table. Codon optimization of heterologous genes is one of the most valuable tools for improving the production level of their corresponding proteins in filamentous fungi. In *A.*

Figures:

Figure 1: Approaches used for improving the production of heterologous proteins in filamentous fungi

Mentioned biomedical entities:

Genes: C3-1, CHY, Don3, ER, Fu, GRAS, GlaA, Kex2, PrtT, UPS, Vib1, cbh1, cbh2, cellulase, glaA, histone, hrdC, sodM

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: *A. niger*, *A. oryzae*, *A. sojae*, *A. terreus*, *A. unguis*, human

Tissues: pulp

Pathways: N/A

Query Result 9/25**The Duality of the MAPK Signaling Pathway in the Control of Metabolic Processes and Cellulase Production in *Trichoderma reesei***

de Paula, Renato Graciano; Antonito, Amanda Cristina Campos; Carraro, Cludia Batista; Lopes, Douglas Christian Borges; Persinoti, Gabriela Felix; Peres, Nalu Teixeira Aguiar; Martinez-Rossi, Nilce Maria; Silva-Rocha, Rafael; Silva, Roberto Nascimento

Scientific Reports

Year: 2018 PMID: 30297963 doi: 10.1038/s41598-018-33383-1 Citations: 18 Score: 11.19

Keywords hits: cellulase: 52, reesei: 66, cbh1: 4, xyr1: 9

Abstract:

In this study, through global transcriptional analysis by RNA-Sequencing, we identified the main changes in gene expression that occurred in two functional mutants of the MAPK genes *tmk1* and *tmk2* in *Trichoderma reesei* during sugarcane bagasse degradation. We found that the proteins encoded by these genes regulated independent processes, sometimes in a cross-talk manner, to modulate gene expression in *T. reesei*. In the *tmk2* strain, growth in sugarcane bagasse modulated the expression of genes involved in carbohydrate metabolism, cell growth and development, and G-protein-coupled receptor-mediated cell signaling. On the other hand, deletion of *tmk1* led to decreased expression of the major genes for cellulases and xylanases. Furthermore, *TMK1* found to be involved in the regulation of the expression of major facilitator superfamily transporters. Our results revealed that the MAPK signaling pathway in *T. reesei* regulates many important processes that allow the fungus to recognize, transport, and metabolize different carbon sources during plant cell wall degradation.

Summary:

The second-most repressed gene in *tmk2* upon growth in sugarcane bagasse encoded a transporter belonging to the MFS permease family, showing 117-fold decrease in expression compared to the parental strain. To identify potential cross-talk between the *TMK1* and *TMK2* signaling in the presence of sugarcane bagasse and glucose, we selected the genes whose expression was differentially regulated in both mutant strains when compared to the parental strain. Deletion of *ime2*, encoding a MAPK in *T. reesei* induced the expression of the three main cellulase genes and repressed the expression of *XYR1* and *CRE1*. Curiously, this gene was also downregulated in *tmk2* in the same condition, suggesting a possible mechanism of cross-talk between the two MAPK genes in *T. reesei*. The transcriptome data of *tmk2* revealed that the MAPK signaling pathway regulates the expression of different genes encoding other signaling pathways during growth in the presence of sugarcane bagasse. For the experiments with sugarcane bagasse, the parental, *tmk2* and *tmk1* strains were grown in glycerol 1% for 24 hours and then transferred to a medium containing sugarcane bagasse. Then, the data of the parental and mutant strains grown on sugarcane bagasse were compared to each other.

Figures:

Figure 1: Expression profile of MAPK genes in the QM6a parental strain grown in sugarcane bagasse and glucose.

Figure 2: Gene Regulatory Network (GRN) of differentially expressed genes in the mutant strains grown in glucose, sugarcane, and glycerol.

Figure 3: Expression pattern of genes commonly expressed between the *tmk1* and *tmk2* representing the number of differentially expressed genes in the presence of glucose and sugarcane bagasse.

Figure 4: Cellulase gene expression profile of *T. reesei* QM6a, *tmk1*, and *tmk2* strains during growth in glucose (gluc), glycerol (glyc), and sugarcane bagasse (SCB).

Figure 5: Holocellulolytic activities of QM6a, *tmk2*, and *tmk1* grown in sugarcane bagasse.

Figure 6: Global view of the MAPK signaling pathway in *tmk2* and *tmk1* mutant strains grown in glucose, glycerol, and sugarcane bagasse.

Figure 7: Heat map of predicted MAPK phosphorylation sites of MFS transporters in the *tmk1* and *tmk2* strains grown in sugarcane bagasse.

Mentioned biomedical entities:

Genes: ABF3, ACE1, ACE2, BGL1, BGL2, CBH1, CBH2, CLP1, CMP1, CRE1, Cellulase, Ctr1, Ctt1, FlibA, Fus3, GNA1, GNA3, GNB1, GNG1, Hog1, LAE1, Log2, MAPK, MAPKKK, MFS, Myb, Sko1, Slt2, Ste11, Ste50, Tec1, XYN2, XYR1, bZIP, bgl1, cbh1, cel6a, cellulase, chk1, myb

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: human

Tissues: N/A

Pathways: N/A

Query Result 10/25

MAPkinases regulate secondary metabolism, sexual development and light dependent cellulase regulation in *Trichoderma reesei*

Schalamun, Miriam; Beier, Sabrina; Hinterdobler, Wolfgang; Wanko, Nicole; Schinnerl, Johann; Brecker, Lothar; Engl, Dorothea Elisa; Schmoll, Monika

Scientific Reports

Year: 2023

PMID: 36732590

doi: 10.1038/s41598-023-28938-w

Citations: 11

Score: 11.18

Keywords hits: cellulase: 48, reesei: 40, cbh1: 15, xyr1: 9

Abstract:

The filamentous fungus *Trichoderma reesei* is a prolific producer of plant cell wall degrading enzymes, which are regulated in response to diverse environmental signals for optimal adaptation, but also produces a wide array of secondary metabolites. Available carbon source and light are the strongest cues currently known to impact secreted enzyme levels and an interplay with regulation of secondary metabolism became increasingly obvious in recent years. While cellulase regulation is already known to be modulated by different mitogen activated protein kinase (MAPK) pathways, the relevance of the light signal, which is transmitted by this pathway in other fungi as well, is still unknown in *T. reesei* as are interconnections to secondary metabolism and chemical communication under mating conditions. Here we show that MAPkinases differentially influence cellulase regulation in light and darkness and that the Hog1 homologue TMK3, but not TMK1 or TMK2 are required for the chemotropic response to glucose in *T. reesei*. Additionally, MAPkinases regulate production of specific secondary metabolites including trichodimerol and bisorbibutenolid, a bioactive compound with cytostatic effect on cancer cells and deterrent effect on larvae, under conditions facilitating mating, which reflects a defect in chemical communication. Strains lacking either of the MAPkinases become female sterile, indicating the conservation of the role of MAPkinases in sexual fertility also in *T. reesei*. In summary, our findings substantiate the previously detected interconnection of cellulase regulation with regulation of secondary metabolism as well as the involvement of MAPkinases in light dependent gene regulation of cellulase and secondary metabolite genes in fungi., b

Summary:

Therefore, we asked whether the regulatory function of the MAPkinases might be connected to the role of VEL1 by testing transcript abundance of *vel1* in deletion strains of *tmk1*, *tmk2* and *tmk3*. Indeed, we found a light dependent regulation of *vel1* in all MAPkinase mutants, with differential impacts either in constant light or in constant darkness. The previously detected only minor effect of TMK2 on *cbh1* transcript abundance may be due to the uncontrolled light conditions during cultivation: Since we observed a clear increase of *cbh1* in *tmk2* in darkness and a decrease in light, what previously was found, may well be a mixture of these effects. In case of TMK3 our results for cellulase regulation are in agreement with previous data, although also here the regulation pattern we observed is more severe, with activity and transcript levels barely detectable anymore. We therefore checked available transcriptome data from comparable conditions for indications if such a feedback might exist, but since we did not find significant regulation of *tmk1*, *tmk2* or *tmk3* in these data, we conclude that this is not the case. Interestingly, in *N. crassa* the OS pathway, corresponding to the Hog1-pathway in yeast and comprising a homologue of TMK3 has no significant influence on cellulase production, which is in contrast to our results. In summary, our data obtained with experiments under controlled light conditions clearly show a light dependent regulatory function of all three MAPkinases on cellulase gene regulation and secreted cellulase activity, which is jeopardized by random light pulses. The GPCR CSG1, which is essential for the chemotropic response of *T. reesei* to glucose, was shown to be required for posttranscriptional regulation of cellulase gene expression. All three mutants were confirmed to only have a single integration of the deletion cassette by copy number determination. All crosses for the analysis of sexual development were performed on 60 mm 2% MEX agar plates at 22 C and 12 h lightdark cycles as previously described. At least three biological replicates were considered in every assay. For NMR spectroscopic measurements -bisorbibutenolide was dissolved in CD3OD and transferred into 5 mm high precision NMR sample tubes.

Figures:

Figure 1: Relevance of MAPkinases for phenotype and biomass formation.

Figure 2: Relevance of MAPkinases for chemotropic response and cellulase regulation.

Figure 3: Relevance of MAPkinases for sorbicillin production and genes involved in secondary metabolism.

Figure 4: Involvement of MAPkinases in mating abilities.

Figure 5: HPLC analysis of MAPK deletion mutants and identification of sorbicillin derivatives.

Figure 6: Schematic representation of the involvement of the MAPkinases, TMK1, TMK2 and TMK3 in sexual development (female fertility), cellulase regulation and secondary metabolism in constant light (LL) and constant darkness (DD).

Mentioned biomedical entities:

Genes: CSG2, CWI, Fus3, Git1, Gpa2, MAPK, MAT1-1, MAT1-2, MPK1, Slt2, Tmk1, Tmk2, VEL1, XYR1, cbh1, cellulase, cre1, env1, peroxidase, pks4, sar1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: *A. flavus*

Tissues: Ascospore, capillary

Pathways: N/A

Query Result 11/25

The -importin KAP8 (Pse1/Kap121) is required for nuclear import of the cellulase transcriptional regulator XYR1, asexual sporulation and stress resistance in *Trichoderma reesei*

Ghassemi, Sara; Lichius, Alexander; Bidard, Frderique; Lemoine, Sophie; Rossignol, Marie-Nolle; Herold, Silvia; Seidl-Seiboth, Verena; Seiboth, Bernhard; Espeso, Eduardo A; Margeot, Antoine; Kubicek, Christian P

Molecular Microbiology

Year: 2015 PMID: 25626518 doi: 10.1111/mmi.12944

Citations: 15

Score: 11.12

Keywords hits: cellulase: 45, reesei: 87, cbh1: 4, xyr1: 55

Abstract:

The ascomycete *Trichoderma reesei* is an industrial producer of cellulolytic and hemicellulolytic enzymes, and serves as a prime model for their genetic regulation. Most of its (hemi-)cellulolytic enzymes are obligatorily dependent on the transcriptional activator XYR1. Here, we investigated the nucleo-cytoplasmic shuttling mechanism that transports XYR1 across the nuclear pore complex. We identified 14 karyopherins in *T.reesei*, of which eight were predicted to be involved in nuclear import, and produced single gene-deletion mutants of all. We found KAP8, an ortholog of *Aspergillus nidulans* Kap1, and *Saccharomyces cerevisiae* Kap121/Pse1, to be essential for nuclear recruitment of GFP-XYR1 and cellulase gene expression. Transformation with the native gene rescued this effect. Transcriptomic analyses of *kap8* revealed that under cellulase-inducing conditions 42 CAZymes, including all cellulases and hemicellulases known to be under XYR1 control, were significantly down-regulated. *kap8* strains were capable of forming fertile fruiting bodies but exhibited strongly reduced conidiation both in light and darkness, and showed enhanced sensitivity towards abiotic stress, including high osmotic pressure, low pH and high temperature. Together, these data underscore the significance of nuclear import of XYR1 in cellulase and hemicellulase gene regulation in *T.reesei*, and identify KAP8 as the major karyopherin required for this process.

Summary:

Using a systematic gene deletion approach, we identified KAP8, the ortholog of Pse1/Kap121 from *S. cerevisiae* and *A. nidulans* Kap1, respectively, to be required for XYR1 import into the nucleus, and demonstrate that this step is essential for cellulase and hemicellulase gene expression. B, expression of *cbh1* in the *kap8* strain was only 2% of that shown by the complemented transformant strain. We therefore concluded that *kap8* is essential for cellulase gene expression and while it cannot be ruled out that other importins can also transport XYR1 into the nucleus is of major importance for the function of XYR1. In order to test whether KAP8 is indeed essential for XYR1 uptake into the nucleus, we expressed a GFP-XYR1 fusion protein in *T. reesei kap8* and its complemented transformant and monitored its subcellular localization under inducing conditions. In *T. reesei*, sporulation of the *kap8* strain was also reduced to 10% of that of the parental strain ($n = 4$ Fig. Table lists all *T. reesei* strains used and produced in this study. The *A. nidulans* strains used in this study are given in Supporting Information Table.

Figures:

Figure 1: Expression of the T .

Figure 2: Nuclear recruitment of GFP-XYR1 was not observed in the *kap8* strain; however, it was restored in the *kap8* complemented transformant (*kap8 ct*).

Figure 3: Number of genes, grouped according to functional similarity, that were induced by sophorose at least fourfold (grey bars) and genes whose sophorose induction was significantly lower (5-fold) in the *kap8* strain compared with the complemented transformant *kap8ct* (white bars).

Figure 4: Growth of A .

Figure 5: Effect of *kap8* knock out on asexual and sexual development in T .

Figure 6: Growth of T .

Figure 7: Inventory of gene groups that were not induced by sophorose but significantly down-regulated (at least 5-fold; $P < 0.05$) in the *kap8* strain compared with the retransformant.

Mentioned biomedical entities:

Genes: ACE1, ACE2, ACE3, Aft1, AlcR, B, CRE1, Cellulase, Gal4, KAP1, KAP3, KAP6, KAP8, Kap104, MAT1-1, MAT1-2, NirA, Nmd5, Pdr1, Pho4, Pse1, Ran, Sln1, Ste12, Sxm1, XYR1, XlnR, Yap1, cbh1, cel7A, cellulase, kap1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: N/A

Tissues: lens, pulp, tube

Pathways: N/A

Query Result 12/25**Ras GTPases Modulate Morphogenesis, Sporulation and Cellulase Gene Expression in the Cellulolytic Fungus *Trichoderma reesei*, bRas Modulate Morphogenesis and Cellulase Formation**

Zhang, Jiwei; Zhang, Yanmei; Zhong, Yaohua; Qu, Yinbo; Wang, Tianhong; Xue, Chaoyang

PLoS ONE

Year: 2012 PMID: 23152805 doi: 10.1371/journal.pone.0048786 Citations: 16 Score: 10.93

Keywords hits: cellulase: 69, reesei: 50, cbh1: 25, xyr1: 47

Abstract:

The model cellulolytic fungus *Trichoderma reesei* (teleomorph *Hypocrea jecorina*) is capable of responding to environmental cues to compete for nutrients in its natural saprophytic habitat despite its genome encodes fewer degradative enzymes. Efficient signalling pathways in perception and interpretation of environmental signals are indispensable in this process. Ras GTPases represent a kind of critical signal proteins involved in signal transduction and regulation of gene expression. In *T. reesei* the genome contains two Ras subfamily small GTPases TrRas1 and TrRas2 homologous to Ras1 and Ras2 from *S. cerevisiae*, but their functions remain unknown.

Summary:

However, detailed functions of these two Ras signal proteins in this organism remain unknown. It is also known that in many fungi, e. g., *S. cerevisiae*, *C. neoformans* and *C. albicans*, Ras plays an important role in activating the cAMP pathway to regulate cell morphology and cell cycle. Taken together, these results suggest that Tr Ras1 is an important signal protein that controls hyphal cell formation and polarized apical growth in *T. reesei* deficiency of the hyphal growth subsequently results in a small and dense colony in the Tr Ras1 deletion mutant Tr Ras1 is involved in controlling asexual development in *T. reesei*. These data suggest that Tr Ras2 may sense the signals except cellulose and act upstream of the transcription regulators to modulate cellulase gene expression. It is believed that the expression of cellulolytic gene is regulated by the cellulase transcription factors in *T. reesei*, brid=pone.0048786-Stricker2 ref-type=bibr13,... Thus, it seems that Tr Ras2 signalling may transmit the light or other signals to modulate cellulase gene expression. It has been discovered that cAMP signalling is involved in regulation of cellulase gene expression,. However, no alteration in the cAMP level on cellulose or glucose is detected in the mutant with a deletion for Tr Ras2, indicating that influence of Tr Ras2 on cellulase gene transcription is independent of cAMP signalling pathway. Xyr1 has been demonstrated as a central transcriptional regulator that controls xylanolytic as well as cellulolytic enzyme genes expression in *T. reesei*, brid=pone.0048786-Stricker2 ref-type=bibr13.

Figures:

Figure 1: Phylogenetic analysis of Ras proteins.

Figure 2: Analysis of transcript levels of TrRas1 and TrRas2 mRNA by qRT-PCR.

Figure 3: Growth of TU-6, TrRas1 and TrRas2 on different carbon sources.

Figure 4: TrRas1 regulates filamentation and sporulation in *T. reesei*.Figure 5: TrRas2 modulates polarized apical growth, branch formation and sporulation in *T. reesei*.Figure 6: TrRas2 influences cellulase formation in *T. reesei*.

Figure 7: Analysis of the influence of TrRas2 on expression of the transcription factors encoding genes.

Figure 8: Influence of Xyr1 overexpression in the TrRas2 deletion strain on regulation of cellulase gene transcription.

Figure 9: TrRas1 and TrRas2 play similar roles in increasing cAMP content.

Figure 10: Schematic model depicting TrRas1/2 signalling in regulating morphogenesis and cellulase gene expression in *T. reesei*.**Mentioned biomedical entities:**

Genes: Ace1, Ace2, Cdc42, Cellulase, Cre1, GNA3, MAP, MAPK, Ras, Ras1, Ras2, RasA, RasB, RasC, Rho, Xyr1, ace1, ace2, actin, cAMP, cbh1, cbh2, cellulase, cre1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, hyphal cell

Organisms: *C. albicans*, *C. neoformans*, human

Tissues: peripheral

Pathways: N/A

Query Result 13/25

The protein methyltransferase TrSAM inhibits cellulase gene expression by interacting with the negative regulator ACE1 in *Trichoderma reesei*

Zhu, Zhihua; Zou, Gen; Chai, Shunxing; Xiao, Meili; Wang, Yinmei; Wang, Pingping; Zhou, Zhihua

Communications Biology

Year: 2024 PMID: 38548869 doi: 10.1038/s42003-024-06072-1 Citations: 4 Score: 10.80

Keywords hits: cellulase: 58, reesei: 48, cbh1: 34, xyr1: 34

Abstract:

Protein methylation is a commonly posttranslational modification of transcriptional regulators to fine-tune protein function, however, whether this regulation strategy participates in the regulation of lignocellulase synthesis and secretion in *Trichoderma reesei* remains unexplored. Here, a putative protein methyltransferase (TrSAM) is screened from a *T. reesei* mutant with the ability to express heterologous α -glucosidase efficiently even under glucose repression. The deletion of its encoding gene *trsam* causes a significant increase of cellulase activities in all tested *T. reesei* strains, including transformants of expressing heterologous genes using *cbh1* promoter. Further investigation confirms that TrSAM interacts with the cellulase negative regulator ACE1 via its amino acid residue Arg383, which causes a decrease in the ACE1-DNA binding affinity. The enzyme activity of a *T. reesei* strain harboring ACE1R383Q increases by 85.8%, whereas that of the strains with *trsam* or *ace1* deletion increases by more than 100%. By contrast, the strain with ACE1R383K shows no difference to the parent strain. Taken together, our results demonstrate that TrSAM plays an important role in regulating the expression of cellulase and heterologous proteins initiated by *cbh1* promoter through interacting with ACE1R383. Elimination and mutation of TrSAM and its downstream ACE1 alleviate the carbon catabolite repression (CCR) in expressing cellulase and heterologous protein in varying degrees. This provides a new solution for the exquisite modification of *T. reesei* chassis.

Summary:

In repressing medium, the pNPGase activities of *atbg-A1*, *atbg-D3* and *atbg-U10* were 5.36 0.69, 121.32 6.87, and 231.32 21.85 IU/mL, respectively, and the pNPGase activity of *Rut-C30* was barely detectable.), which demonstrated that BglS could be expressed when glucose was used as the carbon source in *atbg-D3* and *atbg-U10* strains. To investigate the function of TrSAM in the regulation of cellulase production, interactions between TrSAM and widely known cellulase-related regulators were analyzed using the yeast two-hybrid system. The interaction between TrSAM-AD and ACE1-BD promoted yeast growth in the absence of histidine and adenine and resulted in X-gal activity, whereas the interaction between TrSAM-AD and *Lae1*-BD or other fused regulators did not. It has been testified that ACE1 and XYR1 may bind to the promoters of *xyr1*, *cbh1* and other cellulase or hemicellulase genes competitively. In this study, we found that the methyltransferase TrSAM participates in the repression of cellulase synthesis by the interaction with the negative regulator ACE1 and thus strengthening its binding to the promoter regions of target genes. Although it is possible that the low expression level of ACE1 under inducing condition has weakened its interaction with TrSAM, which further impairs its repressing effect on cellulase expression, the mutation of 383 residue arginine to glutamine or deletion of *trsam* has been demonstrated to improve cellulase production regardless of inducing or repressing conditions (Fig.

Figures:

Figure 1: Evaluation of enzymatic activity and protein production by transformants expressing BglS in inducing and repressing conditions.

Figure 2: Schematic diagram of large missing fragments in *atbg-D3* and *atbg-U10*.

Figure 3: Interaction analysis of protein methyltransferase TrSAM and ACE1.

Figure 4: Evaluation of cellulase production of *Rut-C30* and its mutants.

Figure 5: Comparison of ACE1 and ACE1R383Q abilities to bind DNA.

Figure 6: Schematic model of TrSAM-mediated regulatory mechanism during carbon catabolite repression in *T. reesei*.

Mentioned biomedical entities:

Genes: ACE1, ACE2, BglR, BglS, CBH1, CCR, Cre1, FPA, GAL4, GATA4, GST, Hda1, Lae1, Mig1, MyoD, PRC2, RK, Roc1, Vel1, XYR1, *ace1*, *bglR*, *cbh1*, cellulase, *cre1*, histone, p53

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, protoplast

Organisms: *E. coli*

Tissues: N/A

Pathways: N/A

Query Result 14/25**Enhancement of Cellulase Production in *Trichoderma reesei* via Disruption of Multiple Protease Genes Identified by Comparative Secretomics**

Qian, Yuanchao; Zhong, Lixia; Sun, Yu; Sun, Ningning; Zhang, Lei; Liu, Weifeng; Qu, Yinbo; Zhong, Yaohua

Frontiers in Microbiology

Year: 2019 PMID: 31849916 doi: 10.3389/fmicb.2019.02784 Citations: 14 Score: 10.72

Keywords hits: cellulase: 50, reesei: 55, cbh1: 3, xyr1: 0

Abstract:

The filamentous fungus *Trichoderma reesei* is one of the most studied cellulolytic organisms and the major producer of cellulases for industrial applications. However, undesired degradation of cellulases often happens in culture filtrates and commercial enzyme preparations. Even studies have been reported about describing proteolytic degradation of heterologous proteins in *T. reesei*, there are few systematic explorations concerning the extracellular proteases responsible for degradation of cellulases. In this study, the cellulase activity was observed to rapidly decrease at late cultivation stages using corn steep liquor (CSL) as the nitrogen source in *T. reesei*. It was discovered that this decrease may be caused by proteases. To identify the proteases, comparative secretomics was performed to analyze the concomitant proteases during the cellulase production. 12 candidate proteases from the secretome of *T. reesei* were identified and their encoding genes were individually deleted via homologous recombination. Furthermore, three target proteases (tre81070, tre120998, and tre123234) were simultaneously deleted by one-step genetic transformation. The triple deletion strain P70 showed a 78% decrease in protease activity and a six-fold increase in cellulase activity at late fermentation stages. These results demonstrated the feasibility of improvement of cellulase production by genetically disrupting the potential protease genes to construct the *T. reesei* strains with low extracellular protease secretion. This dataset also provides an efficient approach for strain improvement by precise genetic engineering combined with omics strategy for high-production of industrial enzymes to reduce the cost of lignocellulose bioconversion.

Summary:

This strain with the multiple protease deletions possessed the stable expression levels of endogenous cellulases and thus a more productive *T. reesei* strain suitable for cellulase production. *Trichoderma reesei* QM9414mus53, a hyper-cellulolytic strain with high-efficiency homologous recombination, was kindly provided by and used as the host strain for gene deletion. The colony diameters of the protease gene deletion strains and the parental strain QM9414mus53 were measured at 24, 44, 56, 68, and 76 h after incubation. The activities of Filter paper, EGs and xylanases were measured using Whatman No. Among these proteins, 74 proteins were shared by Q3 and Q7, 36 proteins were unique in Q3, and 18 proteins were specific for Q7. These results further suggested that the three extracellular proteases, tre81070, tre120998, and tre123234, contribute to the major protease activity in *T. reesei* during the fermentation in CPM medium containing CSL as the nitrogen source. Serine proteases have been reported as the key factor for extracellular protein degradation in *T. reesei*. Here, tre123234 was found to be a serine protease at the late stage of fermentation and its gene deletion resulted in a 55% decrease in protease activity, demonstrating that tre123234 is probably the major serine protease related to cellulase degradation in *T. reesei*.

Figures:

Figure 1: Cellulase and protease production by *T. reesei* during cultivation in CPM containing different nitrogen sources: 0.5% (w/v) (NH₄)₂SO₄ (open circle) or 2.0% corn steep liquor (CSL, filled square).

Figure 2: Determination of the extracellular protein and protease activity of *T. reesei*.

Figure 3: The secreted proteins in the 72 h (Q3) and 168 h (Q7) fermentation supernatants of *T. reesei* identified by means of LC-MS/MS.

Figure 4: Evaluation of the ability to produce proteases by the protease gene-deletion strains in *T. reesei*.

Figure 5: Southern blot analysis of the triple protease-gene deletions in *T. reesei* P70.

Figure 6: Production of protease and cellulase by the triple protease gene-deletion strain P70.

Figure 7: The activities of lignocellulolytic component enzymes produced by the triple protease gene-deletion strain P70.

Mentioned biomedical entities:

Genes: Are1, CBH, CBH1, CBH2, CSL, Cel7B, EG1, FPA, TSP1, cellulase

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: N/A

Tissues: seed

Pathways: N/A

Query Result 15/25

Transcriptomic responses of mixed cultures of ascomycete fungi to lignocellulose using dual RNA-seq reveal inter-species antagonism and limited beneficial effects on CAZyme expression

Daly, Paul; van Munster, Jolanda M.; Kokolski, Matthew; Sang, Fei; Blythe, Martin J.; Malla, Sunir; Velasco de Castro Oliveira, Juliana; Goldman, Gustavo H.; Archer, David B.

Fungal Genetics and Biology

Year: 2017 PMID: 27150814 doi: 10.1016/j.fgb.2016.04.005 Citations: 12 Score: 10.69

Keywords hits: cellulase: 3, reesei: 142, cbh1: 2, xyr1: 1

Abstract:

NA

Summary:

The other set of conditions was used to determine whether the enzymes could have any industrial applications and used a buffer and incubation temperature commonly used for saccharification assays. The saccharification assays to quantify glucose, xylose, arabinose and galactose released from straw so as to predict the flux of sugars in the 24 h cultures were performed in 0.5 ml tubes in a total volume of 450 l with 15 mg of washed wheat straw, volumes of the concentrated 24 h culture filtrates equivalent to 1.5 ml of the original culture filtrates, CCMM and sodium azide at a 0.02% w/v final concentration. Sufficient sequencing depth was obtained from each species from mixed cultures samples for statistical analysis comparing transcript abundance amongst the glucose monoculture conditions and mono and mixed species straw culture conditions. There was a positive correlation as measured by the squared Pearson correlation coefficient of 0.74 between the copy number of the 5.8S rRNA region from each species from a mixed culture sample and the number of uniquely mapped reads obtained from each species from the same sample. With regard to genes encoding plant biomass degrading CAZymes, one example is the *T. reesei* -l-arabinofuranosidase which had increased transcript abundance in mixed cultures that included *A. niger* but not in mixed cultures with only *P. chrysogenum* or in the *T. reesei* straw monoculture. 3.4.2 The FPKMs of *P. chrysogenum* CAZy genes from mono and mixed cultures were clustered. The enzyme activities or total protein levels in the 24 h mixed cultures with *T. reesei* were not synergistically greater than in the *T. reesei* straw monocultures, which could be explained by the reduced transcript abundance of the *T. reesei* CAZy genes. To further understand the fungal biology of these interactions in the mixed cultures, we investigated using the RNA-seq data known methods of antagonism or consequences of antagonism. 3.7 Fungal cells are surrounded by cell walls that are important during contact with the environment, including interactions with other species.

Figures:

Figure 1: Activities towards pNP substrates from monoculture time-courses of *A. niger*, *T. reesei* or *P. chrysogenum*.

Figure 2: Monosaccharides (which include inducers of CAZymes) released by the fungi from straw.

Figure 3: The ratio of the 5.8S rRNA from each species present in the two-species mixed cultures.

Figure 4: Summary of the numbers of differentially expressed (DE) genes in comparisons illustrated for (A) *A. niger*, (B) *P. chrysogenum* and (C) *T. reesei*.

Figure 5: Summary of the enrichment pattern of selected GO terms in DE genes from the straw monoculture compared with either the respective glucose control or any mixed species cultures that also contained that species.

Figure 6: Total CAZy transcript abundance in glucose and straw monocultures and mixed species straw cultures for (A) *A. niger*, (B) *T. reesei* and (C) *P. chrysogenum*.

Figure 7: (A) Hierarchical clustering of the *P. chrysogenum* CAZy genes and (B) selected examples of *P. chrysogenum* CAZy genes that have higher abundance in the mixed culture with *T. reesei* and to a lesser extent in other mixed cultures.

Figure 8: (A) Hierarchical clustering of the *T. reesei* CAZy genes and (B) selected examples of *T. reesei* CAZy genes that are members of the indicated clusters.

Figure 9: Saccharification using 4 d culture filtrates.

Mentioned biomedical entities:

Genes: AbaA, B, CE1, CreA, DE, GH, GTs, NRPS, WetA, XYR1, XlnR, arpA, brnA, cellulase, hypB, hypC

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: A. niger, E. coli

Tissues: N/A

Pathways: N/A

Query Result 16/25

Distribution and diversity of enzymes for polysaccharide degradation in fungi

Berlemont, Renaud

Scientific Reports

Year: 2017

PMID: 28302998

doi: 10.1038/s41598-017-00258-w

Citations: 39

Score: 10.67

Keywords hits: cellulase: 43, reesei: 2, cbh1: 0, xyr1: 0

Abstract:

Fungi are important polysaccharide degraders in the environment and for biotechnology. Here, the increasing number of sequenced fungal genomes allowed for systematic identification of genes and proteins involved in polysaccharide degradation in 218 fungi. Globally, 9,003 sequences for glycoside hydrolases and lytic polysaccharide mono-oxygenases targeting cellulose, xylan, and chitin, were identified. Although abundant in most lineages, the distribution of these enzymes is variable even between organisms from the same genus. However, most fungi are generalists possessing several enzymes for polysaccharide deconstruction. Most identified enzymes were small proteins with simple domain organization or eventually consisted of one catalytic domain associated with a non-catalytic accessory domain. Thus unlike bacteria, fungi's ability to degrade polysaccharides relies on apparent redundancy in functional traits and the high frequency of lytic polysaccharide mono-oxygenases, as well as other physiological adaptation such as hyphal growth. Globally, this study provides a comprehensive framework to further identify enzymes for polysaccharide deconstruction in fungal genomes and will help identify new strains and enzymes with potential for biotechnological application.

Summary:

As for many bacterial polysaccharide degraders, fungi were expected to display abundant and diverse sets of proteins and proteins architectures including many multi-domain and multi-activity enzymes. In 218 fully sequenced fungal genomes, 3,607, 1,060, and 2,386 domains for GH targeting cellulose, xylan, and chitin were identified. In addition, the number of identified domains deviated from the number of identified proteins suggesting that some proteins contain several catalytic domains and in some cases some accessory non-catalytic domains. This suggests that both fluctuation in the genome content and the enzymes multi-domain architecture could affect the fungal potential for polysaccharide deconstruction. As of June 2016, the set of publically accessible genomes retrieved from the Myco Cosm portal contained 218 genomes. Beside strains with reduced potential for polysaccharide deconstruction, variation in the frequency of traits of interest suggested that strains from the same class have evolved varying strategies for carbohydrate processing. Next, the genome-specific distribution of traits for polysaccharide processing was investigated. Finally, several proteins domains targeting cellulose, xylan, and chitin were associated with other unexpected catalytic domains. , bFinally, accounting for multi-domain architecture in GHs and LPMOs for the clustering of sequenced fungal genomes, showed a correlation with the clustering based on the distribution of catalytic domain only. As depicted here, the identification of protein domains based on Hidden Markov Profiles provides an easy way to predict the activity of sequenced fungi and to investigate the diversity of enzymes for polysaccharide utilization,.

Figures:

Figure 1: Frequency of GH domains (per 1,000 predicted genes) involved in cellulose, xylan, and chitin deconstruction and LPMO domains in fungal genomes from major classes (numbers in parentheses stand for the number of sequenced genomes).

Figure 2: Conservatism of enzymes for cellulose, xylan, and chitin deconstruction in taxa with more than 1 sequenced genomes, from subphylum to species.

Figure 3: Distribution of domains involved in the deconstruction of cellulose, xylan, and chitin, in sequenced genomes from the subphylum Agaricomycotina.

Figure 4: Distribution of domains involved in the deconstruction of cellulose, xylan, and chitin, in sequenced genomes from the subphylum Pezizomycotina.

Figure 5: Identification of proteins involved in cellulose, xylan, and chitin deconstruction in *Orpinomyces* sp.

Mentioned biomedical entities:

Genes: CBM1, Cellulase, GH, GH18, GHs, cellulase

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: *Agaricus bisporus*, *R. irregularis*

Tissues: N/A

Pathways: N/A

Query Result 17/25**Functional Characterization of Sugar Transporter CRT1 Reveals Differential Roles of Its C-Terminal Region in Sugar Transport and Cellulase Induction in *Trichoderma reesei***

Wang, Zhixing; Yang, Renfei; Lv, Wenhao; Zhang, Weixin; Meng, Xiangfeng; Liu, Weifeng; Atanasova, Lea

Microbiology Spectrum

Year: 2022 PMID: 35852347 doi: 10.1128/spectrum.00872-22 Citations: 7 Score: 10.64

Keywords hits: cellulase: 67, reesei: 18, cbh1: 6, xyr1: 45

Abstract:

The expression of cellulase genes in lignocellulose-degrading fungus *Trichoderma reesei* is induced by insoluble cellulose and its soluble derivatives. Membrane-localized transporter/transceptor proteins have been thought to be involved in nutrient uptake and/or sensing to initiate the subsequent signal transduction during cellulase gene induction. Crt1 is a sugar transporter proven to be essential for cellulase gene induction although the detailed mechanism of Crt1-triggered cellulase induction remains elusive. In this study, we focused on the C-terminus region of Crt1 which is predicted to exist as an unstructured cytoplasmic tail in *T. reesei*. Serial C-terminal truncation of Crt1 revealed that deleting the last half of the C-terminal region of Crt1 hardly affected its transporting activity or ability to mediate the induction of cellulase gene expression. In contrast, removal of the entire C-terminus region eliminated both activities. Of note, Crt1-C5, retaining only the first five amino acids of C-terminus, was found to be capable of transporting lactose but failed to restore cellulase gene induction in the *crt1* strain. Analysis of the cellular localization of Crt1 showed that Crt1 existed both at the plasma membrane and at the periphery of the nucleus although the functional relevance is not clear at present. Finally, we showed that the cellulase production defect of *crt1* was corrected by overexpressing Xyr1, indicating that Xyr1 is a potential regulatory target of the signaling cascade initiated from Crt1.

Summary:

Particularly, while Crt1-C with the removal of the entire C-terminus lost its ability to transport lactose, Crt1-C5 that retains only the first five amino acids of the C-terminus was still transport capable. To further investigate whether the necessity of Crt1 in mediating cellulase induction absolutely depends on its transporting activities, the competence of the constructed Crt1 mutants in inducing cellulase production was evaluated by complementing the *crt1* strain with the wild-type and mutant Crt1 under the control of a copper responsive promoter P_{tcu1}. Remarkable upregulation in the transcription of the main cellulase genes and the key transcriptional activator *xyr1* was detected at 8 h for Re Crt1, Re Crt1-C22, and Re Crt1-CKR strains, which is much earlier than the parental strain TU-6, indicating that constitutive expression of *crt1* facilitated the transcription of cellulase genes on lactose. The induced transcription of cellulase genes was also detected in Re Crt1 strain under Avicel inducing conditions however, the timing of cellulase gene transcription in Re Crt1 was not as advanced as that on lactose (Fig. We found that the transcription of *crt1* is induced by Avicel in a similar pattern with cellulase genes. The transcription of *crt1* was further analyzed in this strain on glucose.

Figures:

Figure 1: Subcellular localization of Crt1-GFP under cellulase induction condition.

Figure 2: Lactose transport activity analysis of wild type Crt1 and its C-terminal mutants.

Figure 3: Functional complementation test of Crt1 C-terminal mutants in the *crt1* strain.Figure 4: Transcriptional analysis of major cellulase genes and *xyr1* in TU-6, *crt1*, and different complementary strains under lactose inducing condition.Figure 5: The carbon catabolite repression on the cellulase gene transcription was not fully relieved by overexpression of *crt1*.Figure 6: Xyr1 is essential but not sufficient for the transcription of *crt1*.Figure 7: Overexpression of Xyr1 in *crt1* bypassed its cellulase expression defect.**Mentioned biomedical entities:**

Genes: ADH1, Ascl, CDT-2, Crt1, EGFR, ER, GLUT1, HXT, LAC12, Mep2, Rgt2, Sspl, Stp1, Xyr1, actin, cbh1, cbh2, cellulase, eg1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: *Escherichia coli*

Tissues: N/A

Pathways: N/A

Query Result 18/25

A short review on SSF an interesting process option for ethanol production from lignocellulosic feedstocks

Olofsson, Kim; Bertilsson, Magnus; Lidn, Gunnar

Biotechnology for Biofuels

Year: 2008

PMID: 18471273

doi: 10.1186/1754-6834-1-7

Citations: 111

Score: 10.55

Keywords hits: cellulase: 17, reesei: 7, cbh1: 0, xyr1: 0

Abstract:

Simultaneous saccharification and fermentation (SSF) is one process option for production of ethanol from lignocellulose. The principal benefits of performing the enzymatic hydrolysis together with the fermentation, instead of in a separate step after the hydrolysis, are the reduced end-product inhibition of the enzymatic hydrolysis, and the reduced investment costs. The principal drawbacks, on the other hand, are the need to find favorable conditions (e.g. temperature and pH) for both the enzymatic hydrolysis and the fermentation and the difficulty to recycle the fermenting organism and the enzymes. To satisfy the first requirement, the temperature is normally kept below 37C, whereas the difficulty to recycle the yeast makes it beneficial to operate with a low yeast concentration and at a high solid loading. In this review, we make a brief overview of recent experimental work and development of SSF using lignocellulosic feedstocks. Significant progress has been made with respect to increasing the substrate loading, decreasing the yeast concentration and co-fermentation of both hexoses and pentoses during SSF. Presently, an SSF process for e.g. wheat straw hydrolyzate can be expected to give final ethanol concentrations close to 40 g L⁻¹ with a yield based on total hexoses and pentoses higher than 70%.

Summary:

In an SHF process, the temperature for the enzymatic hydrolysis can be optimized independently from the fermentation temperature, whereas a compromise must be found in an SSF process. The optimal composition will most certainly depend on the lignocellulosic raw material. The general requirements on an organism to be used in ethanol production is that it should give a high ethanol yield, a high productivity and be able to withstand high ethanol concentrations in order to keep distillation costs low. In addition to these general requirements, inhibitor tolerance, temperature tolerance and the ability to utilize multiple sugars are essential for SSF applications. The optimal enzyme cocktail composition is certainly raw-material specific, and supplementation with extra -glucosidase is as to be expected more important in SHF than in SSF. To decrease the amount of added enzymes needed, investigations of SSF with mixtures of *S. cerevisiae* and the -glucosidase producing yeast strain *Brettanomyces clausenii*, have been undertaken, and compared to SSF with only *S. cerevisiae*. Hence, a higher yeast concentration in the SSF will result in a lower overall ethanol yield if the substrate cost for the production of the yeast is considered.

Figures:

Figure 1: Schematic representation of an SSF process.

Figure 2: Simplified scheme of sugar transport and metabolism in *S. cerevisiae*.

Figure 3: The influence of substrate loading (A), enzyme loading (B), and cell concentration (C) on ethanol yield in SSF of different materials: pretreated barley straw (black circle) 121, pretreated spruce (black diamond) 117, 149, pretreated salix (black square) 120, pretreated willow (black triangle) 122 and pretreated corn stover () 150.

Figure 4: SSF in relation to other process options.

Mentioned biomedical entities:

Genes: CX, Gal2, PDC, XDH, XKS1, XR, adhB, cellulase

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: *C. lusitaniae*, *E. coli*, *Escherichia coli*, *S. cerevisiae*

Tissues: liquid

Pathways: N/A

Query Result 19/25

Metabolic engineering of yeasts by heterologous enzyme production for degradation of cellulose and hemicellulose from biomass: a perspective

Kricka, William; Fitzpatrick, James; Bond, Ursula

Frontiers in Microbiology

Year: 2014 PMID: 24795706 doi: 10.3389/fmicb.2014.00174 Citations: 22 Score: 10.51

Keywords hits: cellulase: 74, reesei: 17, cbh1: 0, xyr1: 0

Abstract:

This review focuses on current approaches to metabolic engineering of ethanologenic yeast species for the production of bioethanol from complex lignocellulose biomass sources. The experimental strategies for the degradation of the cellulose and xylose-components of lignocellulose are reviewed. Limitations to the current approaches are discussed and novel solutions proposed.

Summary:

The most commonly used host for heterologous expression of cellulase genes for CBP is the bakers yeast, *S. cerevisiae*, although the three classes of cellulase genes have also been expressed in other *Saccharomyces* species such as *S. pastorianus* as well as in bacterial species such as *Escherichia coli*. These promoters have been used for heterologous cellulase gene expression in *S. cerevisiae*, however no more than a 2-fold difference in expression was observed in genes driven by these two promoters, with TEF1 generating the highest levels. Another strategy for increasing cellulase activity is to increase the gene copy number. Interesting, very little difference in ethanol yields was observed in yeast strains in which the cellulases were secreted or cell wall tethered, indicating that one approach is not superior to the other. To date, haploid *S. cerevisiae* has been the host of choice for heterologous expression of cellulase genes, although a few studies have examined the expression of cellulase genes in other hosts. Although some success has been achieved using natural xylose fermenting species, their industrial relevance has yet to be demonstrated. Two major xylose utilizing pathways have been identified. Until activity can be increased at this low temperature, it is likely that there will be a need for an enzymatic pre-hydrolysis of biomass prior to fermentation, i. e., SHF in situ.

Figures:

Figure 1: Models for cellulase expression in yeast; heterologous cellulases can be secreted into the medium, tethered individually to the cell wall or assembled into a cellulosome .

Figure 2: Heterologous expression of xylose metabolic pathways in yeast .

Figure 3: Three possible methods for industrial lignocellulose conversion into bioethanol by SFH in situ .

Mentioned biomedical entities:

Genes: BUD21, CBH, CBP, CUP1, Cellulase, GAL, GAL2, GND1, Gal2, HXT1, HXT7, Hxt1, Hxt7, NQM1, PGK1, SUT4, TAL1, TEF1, TKL2, Tn, XDH, XUT3, bgl1, cellulase

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: *A. niger*, *A. oryzae*, *Escherichia coli*, *S. cerevisiae*

Tissues: egg

Pathways: N/A

Query Result 20/25

Design and application of an efficient cellulose-degrading microbial consortium and carboxymethyl cellulase production optimization

Zhang, Guoyan; Dong, Yuanjie

Frontiers in Microbiology

Year: 2022

PMID: 35910619

doi: 10.3389/fmicb.2022.957444

Citations: 9

Score: 10.39

Keywords hits: cellulase: 84, reesei: 6, cbh1: 0, xyr1: 0

Abstract:

Microbial consortia with high cellulase activities can speed up the composting of agricultural wastes with high cellulose contents and promote the beneficial utilization of agricultural wastes. In this paper, rabbit feces and sesame oil cake were used as feedstocks for compost production. Cellulose-degrading microbial strains were isolated from compost samples taken at the different composting stages and screened via Congo red staining and filter paper degradation test. Seven strains, *Trichoderma reesei*, *Escherichia fergusonii*, *Proteus vulgaris*, *Aspergillus glaucus*, *Bacillus mycoides*, *Corynebacterium glutamicum*, and *Serratia marcescens*, with high activities of carboxymethyl cellulase (CMCase), filter paper cellulase (FPase), and -glucosidase (-Gase) were identified and selected for consortium design. Six microbial consortia were designed with these strains. Compared with the other five consortia, consortium VI composed of all seven strains displayed the highest cellulase activities, 141.89, 104.56, and 131.18 U/ml of CMCase, FPase, and -Gase, respectively. The single factor approach and response surface method were employed to optimize CMCase production of consortium VI. The optimized conditions were: culture time 4.25 days, culture temperature 35.5°C, pH 6.6, and inoculum volume 5% (v/v). Under these optimized conditions, the CMCase activity of consortium VI was up to 170.83 U/ml. Fermentation experiment of rabbit feces was carried out by using the consortium VI cultured under the optimal conditions. It was found that the application effect was better than other treatments, and the fermentation efficiency and nutrient content of the pile were significantly improved. This study provides a basis for the design of microbial consortia for the composting of agricultural wastes with high cellulose contents and provides a support for beneficial utilization of agricultural wastes.

Summary:

This indicates the presence of efficient cellulose-degrading microbial strains in the compost pile. In this study, rabbit feces and sesame oil cake were used as feedstocks for compost production, and compost samples were taken at different composting stages for cellulose-degrading microbial strain isolation. Multiple regression was fitted to the experimental data to obtain the optimal values of the factors. The composting experiment was carried out for 26 days in a self-built, aerated static composting boxes, and the composting method was the same as the Section composting and sampling. The mixed rabbit feces had a moisture content at 68.82%, total nitrogen at 21.58 g/kg, TOC at 363.08 g/kg, pH at 6.95, ratio of carbon and nitrogen at 16.83. Sesame oil cake had a moisture content at 11.62%, TN at 69.16 g/kg, TOC at 619.99 g/kg, C/N at 8.97. Compared with EM, the total nitrogen content was increased by 4.12% in CR, C/N ratio was decreased by 21.21%, and the GI was increased by 19.85%. It is of significance to isolate efficient cellulose-degrading indigenous microbial strains and utilize them as a consortium for rapid degradation of cellulose so as to provide technical support for the beneficial utilization of agricultural wastes with high cellulose contents. In this study, seven efficient cellulose-degrading microbial strains were isolated from the compost samples using the Congo red staining test and the filter paper degradation test. Cellulose strains increased the content of active substances in compost, and had better growth promoting effect, which was consistent with the results of this study. In this study, cellulose-degrading microbial strains were isolated from compost samples taken from different composting stages and further screened using the Congo-red staining method and filter paper degradation test.

Figures:

Figure 1: Schematic diagram showing the layout of strain inoculation on plates in the antagonism test.

Figure 2: Photos showing Congo red staining results (A) and clear zone diameters, colony diameters, and ratios of clear zone diameters and colony diameters (B) of some strains.

Figure 3: Activities of cellulase, including carboxymethyl cellulase (CMCase), filter paper cellulase (FPase), and -glucosidase (-Gase), of the seven cellulose-degrading bacterial strains.

Figure 4: Colony and cell morphologies of the seven isolated cellulase-producing strains.

Figure 5: Phylogenetic trees of the seven isolated cellulase-producing microbial strains based on 16S rDNA (bacteria) and D1/D2 domains of the 28S rDNA (fungi) sequences.

Figure 6: Photos showing the results of the antagonism test between the seven cellulase-producing strains.

Figure 7: Optimization of culture conditions for cellulase production by microbial consortium VI by the single factor approach.

Figure 8: Response surface plots showing the interaction effects between factors on carboxymethyl cellulase (CMCase) activity of consortium VI.

Mentioned biomedical entities:

Genes: CK, Cellulase, GA, TN, cellulase

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: A. griseus, Bacillus mycoides, C. glutamicum, Escherichia marmotae, cow, rabbit, rabbits

Tissues: seed

Pathways: N/A

Query Result 21/25

Trichoderma in the light of day Physiology and development

Schmoll, Monika; Esquivel-Naranjo, Edgardo Ulises; Herrera-Estrella, Alfredo

Fungal Genetics and Biology

Year: 2010

PMID: 20466064

doi: 10.1016/j.fgb.2010.04.010

Citations: 54

Score: 10.35

Keywords hits: cellulase: 18, reesei: 21, cbh1: 2, xyr1: 0

Abstract:

In recent years, considerable progress has been made in the elucidation of photoresponses and the mechanisms responsible for their induction in species of the genus *Trichoderma*. Although an influence of light on these fungi had already been reported five decades ago, their response is not limited to photoconidiation. While early studies on the molecular level concentrated on signaling via the secondary messenger cAMP, a more comprehensive scheme is available today. The photoreceptor-orthologs BLR1 and BLR2 are known to mediate almost all known light responses in these fungi and another light-regulatory protein, ENVOY, is suggested to establish the connection between light response and nutrient signaling. As a central regulatory mechanism, this light signaling machinery impacts diverse downstream pathways including vegetative growth, reproduction, carbon and sulfur metabolism, response to oxidative stress and biosynthesis of peptaibols. These responses involve several signaling cascades, for example the heterotrimeric G-protein and MAP-kinase cascades, resulting in an integrated response to environmental conditions.

Summary:

Intriguingly, it was shown that the uptake of sulfate is dependent on the light status, and crucial for growth on cellulose in light and cannot be compensated by addition of the organic sulfur source methionine. Transcription of the sulfur control gene *lim1*, which encodes an E3 ubiquitin ligase, is responsive to light as well as cellulase inducing conditions and regulated by ENVOY. However, the light receptor responsible for the activation of the cAMP-pathway has not been identified. In *T. atroviride* PKA activity has a direct impact not only on photoconidiation, but also on the expression of genes regulated by blue light through the BLR proteins. These results are clear evidence that PKA is involved in light perception, and it has been postulated that the BLR proteins could be a substrate for PKA, or that this kinase could regulate a factor necessary for the transcriptional activity of the BLR complex.^{11,2} In all *Trichoderma* species investigated so far, three G-protein α -subunits and one β -subunit as well as one G-protein γ -subunit have been detected (Schmoll, unpublished data). In addition to the light dependent function of GNA3 in cellulase gene expression in *T. reesei* also a function in peptaibol biosynthesis was observed in *T. atroviride*. Intriguingly, these effects on conidiation were not observed in *T. reesei*.¹² Although considerable progress has been made in elucidating the mechanisms of light response in *Trichoderma* and other fungi, we still lack a comprehensive understanding on how this machinery works and how its function is modulated under different environmental conditions.

Figures:**Mentioned biomedical entities:**

Genes: BLI1, BLR1, BLR2, GNA1, GNA3, LOV, LRE, MAPK, NOP-1, PHR1, PKA, TIMELESS, *blr1*, cAMP, *cbh1*, cellulase, *env1*, *gna1*, *gna3*, *hfb2*, *lim1*, *phr-1*, *vvd*

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: N/A

Tissues: N/A

Pathways: N/A

Query Result 22/25

The promoted hydrolysis effect of cellulase with ultrasound treatment is reflected on the sonicated rather than native brown rice

Li, Hongyan; Xu, Minghao; Yao, Xu; Wen, Yangyang; Lu, Shiyi; Wang, Jing; Sun, Baoguo

Ultrasonics Sonochemistry

Year: 2022 PMID: 35077963 doi: 10.1016/j.ultsonch.2022.105920 Citations: 5 Score: 10.22

Keywords hits: cellulase: 134, reesei: 1, cbh1: 0, xyr1: 0

Abstract:

NA

Summary:

Additionally, the influences of ultrasound on cellulase and its resulting effects on brown rice is unclear and remains to be clarified. In this research, ultrasound intensity and duration were firstly optimized for a food-grade cellulase, with the catalytic activity, kinetics parameters and structure changes of cellulase after ultrasound being analysed. The effect of ultrasound intensity on cellulase activity was evaluated at 0, 0.33, 0.67, 1, 1.33, 1.67, 2, 2.33, 2.67, 3, 3.33 W/cm², respectively, and the effect of treatment duration was also determined. Moreover, brown rice and cellulase solution were co-incubated in the ratio of 1:3 at 40 °C with ultrasonication for 30 min, to study the effect of ultrasound-assisted enzymatic treatment. After each treatment, while the rice solution was filtered and collected, the treated rice was rinsed with deionized water three times, dried in room temperature for 48 h, and then stored in a desiccator for the following various analyses. 2.4.2 SEM was used to examine the changes on surface morphology of brown rice after different treatments. It might be ascribed to that the cracks and fissures on the surface of rice caused by ultrasonication accelerated water diffusion into the internal endosperm during the cooking process, and subsequently led to amylose leaching. Given the remarked impacts of ultrasonication on surface disruption and textural changes of brown rice, the combination of ultrasound and cellulase treatment was further conducted on ultrasonicated rice, aiming to achieve further improvement in the textural quality of brown rice and explore the influence of bran integrity on the effectiveness of cellulase treatment. 3.4 The pre-sonicated brown rice with cellulase treatment possessed the morphological features of both UR and ER, with apparent cracks on the loose surface of rice, as shown in the SEM images. Compared to the previously reported plasma-treated brown rice, significantly more improvement in rice texture was achieved for rice with ultrasound-assisted cellulase treatment, with the hardness being decreased by 33.8% and the stickiness being increased by 86.3%.

Figures:

Figure 1: Ultrasound effects on cellulase activity.

Figure 2: Effects of ultrasonic treatment on structure.

Figure 3: SEM image of brown rice.

Figure 4: Molecular weight distributions of the constituents released from the treated rice.

Figure 5: Textural properties of brown rice.

Figure 6: SEM image of ultrasonicated brown rice.

Figure 7: Textural properties of ultrasonicated brown rice.

Mentioned biomedical entities:

Genes: Bal, Cellulase, ER, amylase, cellulase

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: N/A

Tissues: N/A

Pathways: N/A

Query Result 23/25

Research advances on the consolidated bioprocessing of lignocellulosic biomass

Li, Zhongye; Waghmare, Pankaj Kumar R.; Dijkhuizen, Lubbert; Meng, Xiangfeng; Liu, Weifeng

Engineering Microbiology

Year: 2024 PMID: 39629327 doi: 10.1016/j.engmic.2024.100139 Citations: 5 Score: 10.20

Keywords hits: cellulase: 49, reesei: 44, cbh1: 1, xyr1: 0

Abstract:

Lignocellulosic biomass is an abundant resource for the production of bioproducts. The high cost of cellulase production is the main hurdle for biorefinery industry. CBP is an alternative approach to classical biorefinery for biochemical production. Bacteria, fungi and their co-culture are used for CBP lignocellulosic biorefinery.

Summary:

A wide range of pretreatment techniques, including physical, chemical, and biological pretreatment methods, has been developed to disrupt the structure of lignocellulosic biomass. The developed *S. cerevisiae* CRD5HS strain achieved an ethanol titer of 85.95 and 94.76 g/L from pretreated corn stover and corn cob, respectively, without detoxification or washing the pretreated biomass. *Pichia pastoris* is one of the most commonly used hosts for the eukaryotic expression of heterologous proteins due to its high level of heterologous protein expression, its fast growth rate, and strong pH adaptability. It is also less susceptible to ethanol accumulation and suitable for large-scale high-density fermentation. With a clear division of labor between the two fungi, this co-culture system produced 33.8 g/L of itaconic acid from 270 g/L of -cellulose in a fed-batch fermentation. Fang et al. designed a microbial co-culture system consisting of *T. reesei* C10 and an engineered *S. cerevisiae* strain LGA-1. *T. reesei* C10 could produce more cellulases and thus release more fermentable sugars from the lignocellulose, while the *S. cerevisiae* strain LGA-1 was engineered to metabolize cellobiose for the biosynthesis of D-gluconic acid. The utilization of oxygen by *T. reesei* and *L. pentosus* created a lower redox condition in the co-culture system, for the growth of anaerobic microorganisms, making the co-culture stable. Although CBP is considered a promising strategy for the production of biofuels and biochemicals from lignocellulosic biomass in a single bioreactor, CBP still faces some practical challenges such as the recalcitrant nature of lignocellulosic biomass, the tolerance of strains to the toxic compounds generated during lignocellulosic degradation, the engineering of strains for the high yield of various products, and the complexity of the metabolic interactions between different microorganisms used in consortium.

Figures:

Figure 1: A schematic diagram of the classic three-stage biorefinery process and consolidated bioprocessing technologies.

Mentioned biomedical entities:

Genes: CBH, CBP, EG, FPS, Ldh, NRPs, TAL, bdhB, cbh1, cellulase, pdc, pgk1, ppc

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: *A. succinogenes*, *Bacillus subtilis*, *C. beijerinckii*, *C. cellulovorans*, *C. tyrobutyricum*, *Clostridium acetobutylicum*, *Clostridium beijerinckii* NCIMB 8052, *Clostridium cellulovorans*, *Clostridium tyrobutyricum*, *E. coli*, *Escherichia coli*, *R. toruloides*, *S. cerevisiae*

Tissues: N/A

Pathways: N/A

Query Result 24/25**Innovations in CAZyme gene diversity and its modification for biorefinery applications**

Chettri, Dixita; Verma, Ashwani Kumar; Verma, Anil Kumar

Biotechnology Reports

Year: 2020

PMID: 32963975

doi: 10.1016/j.btre.2020.e00525

Citations: 22

Score: 10.14

Keywords hits: cellulase: 33, reesei: 7, cbh1: 0, xyr1: 0

Abstract:

The review analyses the role of CAZymes in lignocellulose degradation with focus on the different approaches for improving the enzyme activity. Omics based approaches for the analysis of genes, transcripts, proteins and metabolites related to plant degrading mechanism. Directed evolution and rational design as approaches for engineering to enhancing the CAZyme properties. Microbial engineering of the fungal and microbial strains by adapting techniques like CRISPR/cas9 and use of various expression systems. Novel enzymes and cocktail formulation for biorefinery-based application for production of value-added products using low cost plant biomass.

Summary:

This knowledge can be applied for effective production of value-added products. Some important omics technologies that can be applied for prospection of improved CAZymes are discussed. 2.1 For genomic studies of CAZyme producing microorganisms, different traditional molecular fingerprinting techniques are routinely used. The heterologous expression of E1 and GH12 from the bacterium *Acidothermus cellulolyticus* were investigated where both the cellulolytic enzymes were discerned to be expressed successfully by *Zymomonas mobilis* as active and soluble enzymes although E1 was expressed at lower levels with GH12 comprising as much as 4.6 % of total cell proteins. Thus, studies regarding these new class of CAZymes and their genes in plant biomass degrading microbiome can lead to new findings and development of novel CAZymes for their application in the biorefineries. Moreover, additional CAZymes in the form of xylanase, glucuronidase, mannanase etc may be included in the enzyme cocktail for the degradation of the hemicellulose which also adds up to the inaccessibility for cellulose degradation. Protein fingerprinting methods then can be used for study of the newly discovered enzymes for the choice of enzyme cocktail to be used and also to know the drawbacks of enzyme cocktails currently available for their use in the industries. Thus, metagenomics along with screening-based approaches help in the discovery as well as increased knowledge of the synergy between the different CAZymes for efficient biomass degeneration. 5.3 To enhance the economic value of biorefineries, concept of synergy between production of biofuel and value-added product is crucial.

Figures:

Figure 1: Various Omic approaches applied in the field of biology for detailed study of microorganism for generation of potential CAZymes.

Figure 2: The different approaches for CAZyme design.

Figure 3: Ascomycota and Basidiomycota as Microbial Cell Factories (MCFs) for production of wide spectrum of CAZymes.

Figure 4: Rational for formulation of efficient CAZymes Cocktail.

Mentioned biomedical entities:

Genes: CBP, Cas9, CelA, CotA, GH, GHs, GTs, PLs, amylase, cas9, cellulase, pdc, sacB

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: *Bacillus subtilis*, *C. bescii*, *E. coli*, *Escherichia coli*, *S. lividans*

Tissues: pulp

Pathways: N/A

Query Result 25/25**Droplet-based microfluidic platform for high-throughput screening of *Streptomyces***

Tu, Ran; Zhang, Yue; Hua, Erbing; Bai, Likuan; Huang, Huamei; Yun, Kaiyue; Wang, Meng

Communications Biology

Year: 2021

PMID: 34059751

doi: 10.1038/s42003-021-02186-y

Citations: 20

Score: 9.93

Keywords hits: cellulase: 31, reesei: 2, cbh1: 0, xyr1: 0

Abstract:

Streptomyces are one of the most important industrial microorganisms for the production of proteins and small-molecule drugs. Previously reported flow cytometry-based screening methods can only screen spores or protoplasts released from mycelium, which do not represent the filamentous stationary phase *Streptomyces* used in industrial cultivation. Here we show a droplet-based microfluidic platform to facilitate more relevant, reliable and rapid screening of *Streptomyces* mycelium, and achieved an enrichment ratio of up to 334.2. Using this platform, we rapidly characterized a series of native and heterologous constitutive promoters in *Streptomyces lividans* 66 in droplets, and efficiently screened out a set of engineered promoter variants with desired strengths from two synthetic promoter libraries. We also successfully screened out several hyperproducers of cellulases from a random *S. lividans* 66 mutant library, which had 69.2111.4% greater cellulase production than the wild type. Our method provides a fast, simple, and powerful solution for the industrial engineering and screening of *Streptomyces* in more industry-relevant conditions.

Summary:

Consequently, flow rates of 1000 L/h for the oil phase and 650 L/h for the aqueous phase were determined to be optimal and used in further experiments, as these generated 92- μ m droplets at a speed of 443 droplets per second. To develop a droplet-based microfluidic sorting method for *Streptomyces*, we first integrated the constitutive promoter *rps* Lp and an enhanced green-fluorescent protein expression cassette into the *S. lividans* 66 genome, to construct a green fluorescence-positive strain, SLATG. To determine promoter strength, the green fluorescence intensity of each variant in each droplet was measured, and droplets with desired fluorescence intensities were sorted and cultivated for sequencing verification. To address this problem, we demonstrated the application of FADS for the screening and engineering of *S. lividans* 66, to highlight the general application prospects of FADS for the rapid screening of different *Streptomyces* strains. The most important advantage of the FADS-assisted method over the flow cytometry-based high-throughput screening methods of *Streptomyces* is that it enables the screening of *Streptomyces* in its mycelial form, whereas the latter can only screen spores or protoplasts of filamentous microorganisms,,, which do not represent the normal cultivation condition of *Streptomyces*. However, with the rapid advances in the development of small-molecule biosensors and fluorogenic bioassays, the FADS-assisted method could be adapted for engineering and screening *Streptomyces* strains showing enhanced performance in the production of valuable pharmaceuticals and enzymes and other applications. The strains used in this study are listed in Supplementary Data. The resulting 1 mL of spore-containing lactose medium was supplemented with fluorescein-di--cellobioside to a final concentration of 40 M, and then used as the aqueous phase to generate droplets. Origin 2019 was applied for data analysis.

Figures:Figure 1: Preparation and propagation of *S. lividans*66 spores in droplets.Figure 2: Use of FADS for *Streptomyces* detection and screening.Figure 3: Promoter characterization in *S. lividans*66.Figure 4: High-throughput screening of *erm*E and *gapdh*(EL) libraries by FADS.

Figure 5: High-throughput screening of hyperproducers of cellulase by FADS.

Mentioned biomedical entities:

Genes: cellulase

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: *Bacillus subtilis*, *E. coli*, *Escherichia coli*, *Penicillium echinulatum*, *S. lividans*

Tissues: lens, tube

Pathways: N/A